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Combinatorial and Stable Model Categories

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Abstract

We give a unified account of classical model category constructions in a modern way, simplifying many proofs and statements. We then show how to develop the theory of combinatorial and stable model categories in our concise model category framework, such as Jeff Smith's theorem and suspension functors. As an application, we apply the theories to chain complexes and topological spaces and give a new interpretation of their suspension and loop functors; we also conclude by relating model categories to ∞-categories in the end.

Plagiarism statement

The work contained in this thesis is my own work unless otherwise stated.

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Date: June 8, 2026

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1 Introduction

Over the past decades, model categories have proven to be very useful tools in many fields of mathematics. As one of the best frameworks in which one can develop homotopy theories, the theory of model category has gone far beyond algebraic topology, where it originated, and contributed to the areas where homotopy is not even explicit.

An important application of model categories is to the theory of chain complexes. Since the work of [1], mathematicians have recognised that there might be a unified structure behind the derived phenomena in the category of chain complexes of modules over a ring, which led to the research on derived categories such as [2] and on their generalisations called DG-categories such as [3] and [4]. Meanwhile, accumulating study on Koszul duality brings about the coderived category as in [5]; both derived and coderived categories turn out to be concisely described as homotopy categories of some model category. Positselski then unifies the derived, coderived and contraderived categories and describes different types of Koszul dualities in the framework of model categories in [6]; the process is succinctly summarised in [7]. These are the main motivations for devoting a large section to the model category of chain complexes in this paper; we show that by using classical constructions such as cones and cylinders in [1], one can put a nice model structure on chain complexes, obtaining many useful properties of derived categories all at once.

On the other hand, the derived category gives rise to another structure called the triangulated category, as depicted in [1] and [8]. The striking property of having a triangulated homotopy category is not exclusive to the category of chain complexes; it exists in every stable model category. This emphasises the importance of stability of a model category, which originates from the stable phenomena in homotopy theory. Model categories then become the foundation of stable homotopy theories and the study of spectra in [9], [10], and [11]. Additionally, in the framework of model categories, we may produce a stable version of many familiar theorems, such as the stable Dold-Kan correspondence in [12] and [13] and the stable Morita theory in [14]. We will briefly talk about stable model categories in the last section.

Model categories also give a convenient description of simplicial sets and their connection to topological spaces. Research on the model category of simplicial sets leads to the theory of ∞-categories, described in detail in [15]. This paper does not touch on the model category of simplicial sets but depicts the connections between model categories and ∞-categories in several places. The connection is particularly strong when $\mathbf{C}$ is a combinatorial model category due to Dugger's powerful theorem in [16] and [17], in which case the Dwyer-Kan localisation admits a convenient description and the underlying ∞-category of $\mathbf{C}$ can be easily described; see [18] for details. Consequently, combinatorial model categories prove to be important, and Dugger's constructions prove to be universal: Groth describes Dugger's constructions as a left Bousfield localisation, and Dold-Kan correspondence is an example; the same constructions give new theorems in [19]; the derived localisations facilitate theories in [20] and [21]. These are the main reasons why we talk about combinatorial model categories and their consequences.

Motivated by the above, this paper develops theories on model categories, combinatorial model categories, and stable model categories. The first five sections focus on the basics of model categories, combining the accounts in [22], [23], and [24]. Section 6 introduces cofibrantly generated model categories and combinatorial model categories; in particular, we show it suffices to mimic the proof of Hovey's small object argument for proving Jeff Smith's theorem, without the need for further theories on presentability. In section 7, we give a simplified proof of chain complexes being a model category and present the usual model structures on topological spaces. Section 8 briefly discusses stable model categories; in particular, we present a "∞-category version" of the definitions of suspension and loop functors and prove them to be

equivalent to their most common forms, which we then use to determine the functors in the category of chain complexes and topological spaces.

2 Model categories

2.1 Basic constructions

Before properly defining model categories, we make some preliminary definitions. Given a category $\mathbf{C}$, we denote by $\mathrm{Mor}(\mathbf{C})$ the category whose objects are morphisms of $\mathbf{C}$ and whose morphisms are commutative squares.

Definition 2.1 (Retracts)**.** A morphism f in $\mathbf{C}$ is a *retract* of another morphism g if there is a commutative square

$$\begin{array}{ccccc} A & \longrightarrow & C & \longrightarrow & A \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} \\ B & \longrightarrow & D & \longrightarrow & B \end{array}$$

where the horizontal composites are identities.

Definition 2.2 (Lifting properties)**.** Let $i : A \to B$, $p : X \to Y$ be morphisms in $\mathbf{C}$. Then i has the *left lifting property* (abbreviation: LLP) with respect to p, and p has the *right lifting property* (abbreviation: RLP) with respect to i if, for every commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle p} \\ B & \xrightarrow{g} & Y \end{array}$$

there exists a lift $h : B \to X$ such that $hi = f$ and $ph = g$.

Definition 2.3 (Model category)**.** A *model category* is a category $\mathbf{C}$ with three distinguished classes of maps:

(i) *weak equivalences*

(ii) *cofibrations*

(iii) *fibrations*, closed under composition and containing all the identity maps. We call a map that is both a weak equivalence and a cofibration an *acyclic* (or a *trivial*) *cofibration*, and we call a map that is both a weak equivalence and a fibration an *acyclic* (or a *trivial*) *fibration* (we will use acyclic and trivial interchangeably). The collections of maps satisfy:

MC1 Limits and colimits (initial object, coproducts and pushouts!) exist in $\mathbf{C}$.

MC2 (*Two out of three*) If f and g are morphisms of $\mathbf{C}$ such that two of f, g and gf are weak equivalences, then so is the third.

MC3 (*Retracts*) If f and g are morphisms of $\mathbf{C}$ such that f is a retract of g and g is a weak equivalence, cofibration, or fibration, then so is f.

MC4 (*Lifting*) Acyclic cofibrations have the left lifting property with respect to fibrations, and cofibrations have the left lifting property with respect to acyclic fibrations.

MC5 (*Functorial factorisation*) There exist functors $\alpha, \beta : \mathrm{Mor}(\mathbf{C}) \to \mathrm{Mor}(\mathbf{C})$ such that for any morphism f, $f = \beta(f)\alpha(f)$, where $\alpha(f)$ is a cofibration and $\beta(f)$ is an acyclic fibration; there also exist functors $\delta, \gamma : \mathrm{Mor}(\mathbf{C}) \to \mathrm{Mor}(\mathbf{C})$ such that for any morphism f, $f = \gamma(f)\delta(f)$, where $\delta(f)$ is an acyclic cofibration and $\gamma(f)$ is a fibration.

In particular, **MC5** implies for any f' isomorphic to f, the "components" $\alpha(f'), \beta(f')$ remain isomorphic. Some authors do not assume the factorisation to be functorial, but there is no harm in imposing this restriction; see [24] for details.

By **MC1**, an initial object $\emptyset$ and a terminal object $*$ exist in $\mathbf{C}$, giving rise to the following definition:

Definition 2.4 (Cofibrant and fibrant objects)**.** Let A and X be objects of $\mathbf{C}$. A is cofibrant if $\emptyset \to A$ is; X is fibrant if $X \to *$ is.

Example 2.5 ($\mathbf{C}^{\mathrm{op}}$ and under category). If $\mathbf{C}$ is a model category, $\mathbf{C}^{\mathrm{op}}$ is also a model category with f^{op} being:

(i) a weak equivalence if f is a weak equivalence,

(ii) a cofibration if f is a fibration,

(iii) a fibration if f is a cofibration.

For $A \in \mathrm{Obj}(\mathbf{C})$, the under category $A{\downarrow}\mathbf{C}$ is the category where an object is a map $f : A \to X$ and a morphism between $f : A \to X$ and $g : A \to Y$ is $h : X \to Y$ such that $hf = g$. $A{\downarrow}\mathbf{C}$ is also a model category with $h : (A \to X) \to (A \to Y)$ being:

(i) a weak equivalence if $h : X \to Y$ is a weak equivalence,

(ii) a cofibration if $h : X \to Y$ is a cofibration,

(iii) a fibration if $h : X \to Y$ is a fibration.

Remark. An example of an under category is $*{\downarrow}\mathbf{Top}$, the category of pointed spaces. We denote it as $\mathbf{Top}_*$ with objects of the form $(X, *) := (* \to X)$.

Proposition 2.6 (Duality)**.** *By investigating the opposite category, we obtain: if P is a true statement about all model categories, then the statement P^* obtained by reversing the arrows and switching "cofibrations" and "fibrations" is also true.*

2.2 Lifting Properties and Pushouts

The following techniques are very common to create new maps and objects:

(i) given $f : A \to X$, $p : X' \to X$, we would like to "lift" the map f to $f' : A \to X'$;

(ii) given $f : A \to X$, $g : A \to Y$, we would like to extend f and g so that they map to a "smallest" common object (for example the cylinder).

By **MC4**, the first technique requires p to be a fibration, and i precomposing f to be a (acyclic) cofibration; the new object coming from the second technique additionally inherits some nice properties from f and g.

Proposition 2.7 (LLP determines cofibrations)**.** *Let $\mathbf{C}$ be a model category.*

(i) *The cofibrations (resp. acyclic cofibrations) in $\mathbf{C}$ are the maps having LLP with respect to acyclic fibrations (resp. fibrations).*

(ii) *The fibrations (resp. acyclic fibrations) in $\mathbf{C}$ are the maps having RLP with respect to acyclic cofibrations (resp. cofibrations).*

Proof. Suppose $f : K \to L$ has LLP with respect to all acyclic fibrations. **MC5** factorises f as $K \hookrightarrow L' \xrightarrow{\sim} L$, and by assumption there exists a lifting $g : L \to L'$ in:

$$\begin{array}{ccc} K & \xrightarrow{i} & L' \\ {\scriptstyle f}\downarrow & & {\scriptstyle p}\downarrow{\scriptstyle \sim} \\ L & \xrightarrow[\mathrm{id}]{} & L \end{array}$$

This implies f is a retract of i:

$$\begin{array}{ccccc} K & \xrightarrow{\mathrm{id}} & K & \xrightarrow{\mathrm{id}} & K \\ {\scriptstyle f}\downarrow & & {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ L & \xrightarrow[g]{} & L' & \xrightarrow[p]{} & L \end{array}$$

The idea is that if f has LLP, then by factorisation the identity lifts to a map, which makes f a retract of a cofibration i.

□

Remark. The proposition implies that for a model category, we only need to give fibrations (resp. cofibrations) and weak equivalences, and cofibrations (resp. fibrations) are naturally defined to be maps having LLP with acyclic fibrations (resp. acyclic cofibrations).

Proposition 2.8 (Pushouts inherit (co)fibrations)**.** **C** *is a model category.*

(i) *The class of (acyclic) cofibrations is stable under cobase change.*

(ii) *The class of (acyclic) fibrations is stable under base change.*

Proof. The idea is to show that "LLP with respect to acyclic fibrations" is inherited from i in a pushout diagram. Consider the following pushout digram, where i is a cofibration:

$$\begin{array}{ccc} K & \xrightarrow{f} & K' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ L & \xrightarrow[g]{} & L' \end{array}$$

Adjoining this with the lifting problem of j, we obtain:

$$\begin{array}{ccccc} K & \xrightarrow{f} & K' & \xrightarrow{a} & A \\ {\scriptstyle i}\downarrow & & {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle p} \\ L & \xrightarrow[g]{} & L' & \xrightarrow[b]{} & B \end{array}$$

By LLP of i, there is a lifting $h : L \to A$, and by universal property of pushouts, h and a induce the desired map $h' : L' \to A$.

□

3 Homotopy

3.1 Cylinder objects

Definition 3.1. A cylinder object for A is an object $A \wedge I$ of $\mathbf{C}$ together with a diagram:

$$A \overset{\mathrm{in}_0}{\to} A \coprod A \overset{i}{\to} A \wedge I \overset{\sim}{\to} A$$

$$A \overset{\mathrm{in}_1}{\to} A \coprod A \overset{i}{\to} A \wedge I \overset{\sim}{\to} A$$

where $i_0 := i \cdot \mathrm{in}_0$, $i_1 := i \cdot \mathrm{in}_1$, and the overall composition is id_A.

The cylinder object "factorises" $\mathrm{id}_A + \mathrm{id}_A : A \coprod A \to A$, and it's called

(i) a *good cylinder object*, if $A \coprod A \overset{i}{\to} A \wedge I$ is a cofibration, and

(ii) a *very good cylinder object*, if in addition $A \wedge I \to A$ is a fibration.

Remark. If there is a cylinder object for A, then there is a *very good cylinder object* for A, by first factorising i (**MC5**) to obtain a good cylinder object $A \wedge I'$, and again factorising $A \wedge I' \to A$ to obtain a very good cylinder object. Note that we are using the fact that the three classes are closed under composition.

Also, by **MC2**, i_0, i_1 are also weak equivalences since id_A and $A \wedge I \to A$ are.

Remark. $A \times [0,1]$ is a (not usually good) cylinder object for $A \in \mathrm{Obj}(\mathbf{Top})$, although there is a sense of "embedding" two disjoint spaces A into the cylinder.

Lemma 3.2. *If A is cofibrant and $A \wedge I$ good, then $i_0, i_1 : A \to A \wedge I$ are acyclic cofibrations.*

Proof. The key is that $A \coprod A$ is defined by the following pushout diagram:

$$\begin{array}{ccc} \emptyset & \longrightarrow & A \\ {\scriptstyle \text{cofibration}}\downarrow & & \downarrow{\scriptstyle \mathrm{in}_0} \\ A & \xrightarrow{\mathrm{in}_1} & A \coprod A \end{array}$$

From in_0 is a cofibration, and i_0 is also a cofibration. □

3.2 Left homotopy

Definition 3.3. $f, g : A \to X$ is *left homotopic* ($f \overset{l}{\sim} g$) if there exists $H : A \wedge I \to X$

$$A \coprod A \overset{i_0 + i_1}{\to} A \wedge I \overset{H}{\to} X$$

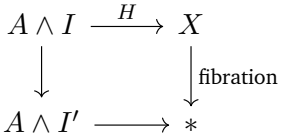
such that $H(i_0 + i_1) = f + g$.

The left homotopy H is thus a map from $A \wedge I$ such that $f = Hi_0$, $g = Hi_1$. It's good (resp. very good) if $A \wedge I$ is good (resp. very good); it's then clear that there always exists a good left homotopy between two left homotopic maps.

Remark. $f \overset{l}{\sim} g$, then f weak equivalence iff g is, transmitted through H by **MC2**.

Lemma 3.4. *If X is fibrant and $f \overset{l}{\sim} g$, then there exists a very good left homotopy between f and g.*

Proof. The key is to "lift" $H : A \wedge I \to X$ to a "bigger" domain ($A \wedge I \xhookrightarrow{\sim} A \wedge I'$):



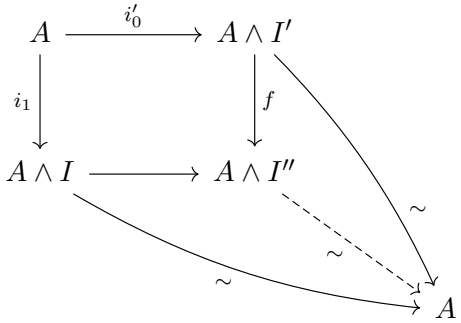
Applying **MC4** gives $H' : A \wedge I' \to X$.

□

Lemma 3.5. *If A is cofibrant, then $\overset{l}{\sim}$ is an equivalence relation on* $\mathrm{Hom}_C(A, X)$.

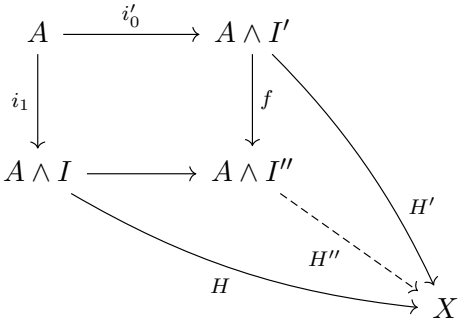
Proof. The first key fact is we can choose the cylinder object and the coproduct. Choose A to be the cylinder for A, and then f is a left homotopy between f and f. Choose the target of $\mathrm{in}_1 + \mathrm{in}_0 : A \coprod A \to A \coprod A$ to be the coproduct (switching factors), and then H is still the left homotopy between g and f.

Now suppose $f \overset{l}{\sim} g$ and $g \overset{l}{\sim} h$; choose good left homotopy $H : A \wedge I \to X$ from f to g and $H' : A \wedge I' \to X$ from g to h. The key is to define $A \wedge I''$ to be the pushout to the following diagram:



which shows $A \wedge I''$ is indeed a cylinder: 2.8 and 3.2 shows f is acyclic cofibration (weak equivalence is inherited *along with* cofibration!), and commutativity plus **MC2** shows $A \wedge I''$ is weak equivalence.

Pushing out H' and H also give H'' (they both compopse to g!):



H'' is indeed a homotopy from f to h by composing with i_0 and i'_1.

□

Remark. Let $\pi^l(A, X)$ denote the set of equivalences classes of $\mathrm{Hom}_{\mathbf{C}}(A, X)$ under the equivalence relation generated by left homotopy.

Here "generated" means that $f \sim g$ if $fR^{\pm 1}f_1R^{\pm 1}\dots R^{\pm 1}f_nR^{\pm 1}g$, where $f_iR^{\pm 1}f_{i+1}$ is $f_i \overset{l}{\sim} f_{i+1}$ or $f_{i+1} \overset{l}{\sim} f_i$.

We then observe how $p : X \to X'$ and $h : A' \to A$ can tweak the set $\pi^l(A, X)$. Under certain constraints, it turns out composing p and precomposing h both respect the homotopy, and acyclic fibration p can induce an isomorphism to $\pi^l(A, X')$.

Lemma 3.6. *A is cofibrant. Acyclic fibration $p : Y \to X$ induces a bijection:*

$$p_* : \pi^l(A, Y) \to \pi^l(A, X), \quad [f] \mapsto [pf]$$

Proof. Well-definedness follows from composing p. The key idea is to lift $g : A \to X$ to $f : A \to Y$ (onto) and to lift $H : A \wedge I \to X$ to $H' : A \wedge I \to Y$ (one-to-one).

For $[g] \in \pi^l(A, X)$,

$$\begin{array}{ccc} \emptyset & \longrightarrow & Y \\ \downarrow & & \downarrow{\scriptstyle p}\sim \\ A & \xrightarrow{g} & X \end{array}$$

gives $f : A \to Y$ directly, $[g] = p_*([f])$.

For homotopy H between $pf \overset{l}{\sim} pg$, $f, g : A \to Y$:

$$\begin{array}{ccc} A \coprod A & \xrightarrow{f+g} & Y \\ \downarrow & & \downarrow{\scriptstyle p}\sim \\ A \wedge I & \xrightarrow{H} & X \end{array}$$

gives $H' : A \wedge I \to Y$, which is a homotopy by commutativity: $f \overset{l}{\sim} g$.

□

Lemma 3.7. *X is fibrant. Then for $h : X' \to X$, $f \overset{l}{\sim} g$, we have $fh \overset{l}{\sim} fh$.*

More generally, there is a map:

$$\pi^l(A', A) \times \pi^l(A, X) \to \pi^l(A', X), \quad ([h], [f]) \mapsto [fh].$$

Proof. By 3.4, choose a very good homotopy $H : A \wedge I \to X$; choose a good cylinder $A' \wedge I$ for A'. The key is to find a map to precompose: $k : A' \wedge I \to A \wedge I$. It's done by considering the commutative diagram formed by two cylinders:

$$\begin{array}{ccccc} A' \coprod A' & \xrightarrow{h \coprod h} & A \coprod A & \xrightarrow{i} & A \wedge I \\ {\scriptstyle j}\downarrow & & & & \downarrow\sim \\ A' \wedge I & \xrightarrow[\sim]{} & A' & \xrightarrow[h]{} & A \end{array}$$

The "good" and "very good" give cofibrations and fibrations, producing k. Hk is the desired homotopy. □

3.3 Path objects and right homotopies

Definition 3.8. A path object for X is an object X^I of $\mathbf{C}$ together with a diagram:

$$X \xrightarrow{\sim} X^I \xrightarrow{p} X \times X \xrightarrow{\mathrm{pr}_0} X$$

$$X \xrightarrow{\sim} X^I \xrightarrow{p} X \times X \xrightarrow{\mathrm{pr}_1} X$$

where $p_0 := \mathrm{pr}_0 \cdot p$, $p_1 := \mathrm{pr}_1 \cdot p$, and the overall composition is id_X.

The cylinder object "factorises" $(\mathrm{id}_X, \mathrm{id}_X) : X \to X \times X$.

Definition 3.9. $f, g : A \to X$ is *right homotopic* ($f \overset{r}{\sim} g$) if there exists $H : A \to X^I$

$$A \xrightarrow{H} X^I \xrightarrow{(p_0,p_1)} X$$

such that $(p_0, p_1)H = (f, g)$. Hence we have $f = p_0 H$, $g = p_1 H$.

Proposition 3.10. *The dual version in terms of path objects and right homotopy is:*

(i) *X fibrant, X^I good path object, then p_0, p_1 are fibrations.*
This implies $\overset{r}{\sim}$ is an equivalence relation on $\mathrm{Hom}_{\mathbf{C}}(A, X)$.

(ii) *A cofibrant, then there is a very good right homotopy from f to $g : A \to X$.*
This implies $hf \overset{r}{\sim} hg$ for $f \overset{r}{\sim} g$ and $h : X \to Y$. More generally, there is a map $\pi^r(A, X) \times \pi^r(X, Y) \to \pi^r(A, Y)$.

(iii) *X fibrant, $i : A \to B$ acyclic cofibration, then precomposition with i induces a bijection:*

$$i^* : \pi^r(B, X) \to \pi^r(A, X)$$

Remark. Right homotopy works well with precomposition.

3.4 Homotopy

Lemma 3.11. *If A is cofibrant and $f \overset{l}{\sim} g$, then $f \overset{r}{\sim} g$; if X is fibrant and $f \overset{r}{\sim} g$, then $f \overset{l}{\sim} g$.*

Proof. Given a left homotopy H, we view it not only as a factorisation of f, but also a parallel map with f:

$$A \xrightarrow{i_0} A \coprod A \xrightarrow{i_0+i_1} A \wedge I \xrightarrow{j} A \xrightarrow{f} X$$

where $H : A \wedge I \to X$. Choose a good path object factorising $f : A \to X$:

$$X \xrightarrow{q} X^I \xrightarrow{(p_0,p_1)} X \times X$$

The ultimate goal is to "lift" $H : A \wedge I \to X$ to $H' : A \to X^I$, and we first need to produce a map into $X \times X$ by joining f and H together:

$$\begin{array}{ccc} A & \xrightarrow{qf} & X^I \\ \downarrow{\scriptstyle i_0} & & \downarrow{\scriptstyle (p_0,p_1)} \\ A \wedge I & \xrightarrow{(fj,H)} & X \times X \end{array}$$

Cofibrant A implies i_0 is a cofibration, and this gives a lift $K : A \wedge I \to X^I$ (by using f-side of the homotopy). $H' = Ki_1$ is then the desired right homotopy (note Ki_1 is homotopic to qf through K). □

Remark. Assume A is cofibrant and X is fibrant. The proof shows that, given any fixed cylinder and path, if $f \sim g$, it is necessary that $f \overset{r}{\sim} g$ through the given path object (since we get $f \overset{l}{\sim} g$, and constructing as in the proof gives the result).

We say $f \sim g$ if $f \overset{l}{\sim} g$ and $f \overset{r}{\sim} g$, and if A is cofibrant, X is fibrant, then we denote $\pi(A, X) := \pi^l(A, X) = \pi^r(A, X)$.

Lemma 3.12 (Weak equivalence $\Leftrightarrow$ Homotopy inverse)**.** *$f : A \to X$ where A and X are both fibrant and cofibrant. Then f is a weak equivalence iff there exists $g : X \to A$ such that gf and fg are homotopic to the respective identity maps.*

Proof. First assume f is a weak equivalence. The key idea is to factorise f, find a one-sided inverse for its components, and show it's a homotopy inverse via the bijection. We can factor f as:

$$A \overset{q}{\underset{\sim}{\hookrightarrow}} C \overset{p}{\underset{\sim}{\twoheadrightarrow}} X$$

q cofibrant into X fibrant always gives $r : C \to A$ in the following diagram:

$$\begin{array}{ccc} A & \xrightarrow{q} & A \\ \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \text{cofib.}} \\ C & \longrightarrow & * \end{array}$$

Then, acyclic cofibration q gives a bijection $q^* : \pi^r(C, C) \to \pi^r(A, C)$, and the inverse trick gives:

$$q^*([qr]) = [qrq] = [q] = q^*(\mathrm{id}_C) \implies qr \sim \mathrm{id}_C.$$

Thus $f = pq$ has a homotopy inverse rs, where s is similarly a homotopy inverse of p.

Next, assume f has a homotopy inverse. We again factorise $f = pq$, q acyclic cofibration thus having an inverse r. The key idea is to try finding an object closest to p such that it is *homotopic to a weak equivalence* (see remark under 3.3), and passing this to p through retraction.

By 3.7 and 3.10, homotopy preserves compositions. $f = pq \implies fr \sim p$, if we can find some object $s \sim qg$, then everything works out (via $gf \sim \mathrm{id}_A$). We then use the other homotopy $fg \sim \mathrm{id}_X$, say H, to construct such s:

$$\begin{array}{ccc} X & \xrightarrow{qg} & C \\ \downarrow{\scriptstyle i_0} & & \downarrow{\scriptstyle p} \\ X \wedge I & \xrightarrow{H} & X \end{array}$$

Then there is a lift $H' : X \wedge I \to C$, and let $s = H' i_1$. Similar to the proof of 3.11, $s \sim qg$ by the homotopy H', which implies $sp \sim sfr \sim qgfr \sim \mathrm{id}_C$. Additionally, from the diagram

$$\begin{array}{ccccc} C & \xrightarrow{\mathrm{id}_C} & C & \xrightarrow{\mathrm{id}_C} & C \\ {\scriptstyle p}\downarrow & & {\scriptstyle sp}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow[s]{} & C & \xrightarrow[p]{} & X \end{array}$$

p is the retract of sp, and thus p is a weak equivalence.

□

4 Homotopy category

4.1 Constructions

To talk about homotopy of maps between any X and Y, we need a formal way to associate to each of them an object that is both fibrant and cofibrant. Specifically, this involves two functors Q and R, defined on the categories below.

Definition 4.1. $\pi\mathbf{C}_c/\pi\mathbf{C}_f/\pi\mathbf{C}_{cf}$ is the category of cofibrant/fibrant/cofibrant-fibrant objects whose morphisms are right homotopy/left homotopy/homotopy classes of maps.

For each object X and each object Y, we can define functors $Q : \mathbf{C} \to \pi\mathbf{C}_c$ and $R : \mathbf{C} \to \pi\mathbf{C}_f$, associating objects QX and RY by **MC5** as follows:

$$\emptyset \to QX \xrightarrow{\sim} X, \quad Y \xhookrightarrow{\sim} RY \to *$$

where $p_X : QX \xrightarrow{\sim} X, i_Y : Y \xhookrightarrow{\sim} RY$.

Remark. Well-definedness of the subcategories is ensured by 3.7. Morphisms induced by Q and R are stated in the following lemma (we specify $QX = X$ if X is cofibrant, $RY = Y$ if Y is fibrant).

Lemma 4.2. *Given $f : X \to Y$, there exists a map $\tilde{f} : QX \to QY$:*

$$\begin{array}{ccc} QX & \xrightarrow{\tilde{f}} & QY \\ \downarrow{\scriptstyle p_X} & & \downarrow{\scriptstyle p_Y} \\ X & \xrightarrow{f} & Y \end{array}$$

The map $\tilde{f}$ depends up to homotopy on f, and is a weak equivalence if and only if f is. If Y is fibrant, then $\tilde{f}$ depends up to homotopy on the left homotopy class of f.

There is also a map $\bar{f} : RX \to RY$, which (if X is cofibrant) depends up to homotopy on (the right homotopy class of) f.

Proof. Naturally we have a diagram:

$$\begin{array}{ccc} \emptyset & \longrightarrow & QY \\ \downarrow & & \downarrow{\scriptstyle p_Y} \\ QX & \xrightarrow{f \cdot p_Y} & Y \end{array}$$

MC4 gives a lift $\tilde{f}$ such that $p_Y \cdot \tilde{f} = f \cdot p_X$. Then:

- By 3.6, p_Y is bijective with respect to left homotopy $\implies$ $\tilde{f}$ is unique up to left homotopy.
- By 3.11, left homotopic maps from cofibrant QX are also right homotopic $\implies$ unique up to homotopy.
- By **MC2**, weak equivalence holds.
- By 3.7, left homotopy of f doesn't change the left homotopy class of $f \cdot p_Y$ if Y fibrant.

□

Remark. The functors Q and R indeed preserve the identity and composition by the uniqueness property in the lemma.

The last statements in 4.2 clearly gives us the restricted functors Q' and R' (it suffices to prove the case when the maps are directly related), as stated in the following lemma:

Lemma 4.3. *The restriction of Q gives $Q' : \pi\mathbf{C}_f \to \pi\mathbf{C}_{cf}$; the restriction of R gives $R' : \pi\mathbf{C}_c \to \pi\mathbf{C}_{cf}$.*

Definition 4.4. The homotopy category $\mathrm{Ho}(\mathbf{C})$ of a model category $\mathbf{C}$ is the category with the same objects as $\mathbf{C}$ and with

$$\mathrm{Hom}_{\mathrm{Ho}(\mathbf{C})}(X, Y) = \mathrm{Hom}_{\pi\mathbf{C}_{cf}}(R'QX, R'QY) = \pi(RQX, RQY).$$

There is a functor $\gamma : \mathbf{C} \to \mathrm{Ho}(\mathbf{C})$ keeping the same objects and sending $f \in \mathrm{Hom}_{\mathbf{C}}(X, Y)$ to $R'Q(f) \in \mathrm{Hom}_{\pi\mathbf{C}_{cf}}(R'QX, R'QY)$.

Remark. If X and Y are both fibrant and cofibrant, then γ is clearly surjective (just take the preimage of f to be f itself) and induces a bijection $\pi(X, Y) \cong \mathrm{Hom}_{\mathrm{Ho}(\mathbf{C})}(X, Y)$ (by putting f in equivalence classes).

4.2 Properties

The following proposition states two important properties of $\mathrm{Ho}(\mathbf{C})$, highlighting how weak equivalences determine the isomorphisms and thus the whole $\mathrm{Ho}(\mathbf{C})$ (to be formalised later).

Proposition 4.5. *If f is a morphism in $\mathbf{C}$, then $\gamma(f)$ is an isomorphism in $\mathrm{Ho}(\mathbf{C})$ if and only if it is a weak equivalence. The morphisms in $\mathrm{Ho}(\mathbf{C})$ are generated by, under γ, the images of morphisms in $\mathbf{C}$ and the inverses of the images of weak equivalences in $\mathbf{C}$.*

Proof. 3.12 and 4.2 (that Q and R preserve weak equivalences) directly give the first result.

From the above, for weak equivalences $p_X : QX \to X$, $i_{QX} : QX \to RQX$, $\gamma(p_X)$ and $\gamma(i_{QX})$ are isomorphisms, and $\gamma(i_{QX})\gamma(p_X)^{-1}$ is an isomorphism from $RQX \to X$ in $\mathrm{Ho}(\mathbf{C})$. For any $f : X \to Y$ in $\mathrm{Ho}(\mathbf{C})$, precomposing f with it, and similarly on the other side, gives us $g : RQX \to RQY$. Then by 4.1, $g = \gamma(f')$ for some $f' \in \mathrm{Hom}_{\mathbf{C}}(RQX, RQY)$, and organising the terms gives:

$$f = \gamma(p_Y)\gamma(i_{QY})^{-1}\gamma(f')\gamma(i_{QX})\gamma(p_X)^{-1}$$

.

□

The following consequence says natural transformations between functors $\mathrm{Ho}(\mathbf{C}) \to \mathbf{D}$ are determined by their behavior on $\mathbf{C}$.

Corollary 4.6. *If F and G are two functors $\mathrm{Ho}(\mathbf{C}) \to \mathbf{D}$ and $t : F\gamma \to G\gamma$ is a natural transformation, then t also gives a natural transformation from F to G.*

Proof. It suffices to check that for each morphism h in $\mathrm{Ho}(\mathbf{C})$, an appropriate diagram $D(h)$ commutes (D here only denotes the diagram). $D(h)$ commutes for $h = \gamma(f)$ or $h = \gamma(g)^{-1}$ for morphism f or weak equivalence g (we know $D(\gamma(g))$ commutes, and composing it with $D(\gamma(g)^{-1})$ we have $D(\mathrm{id})$ commutes, thereby showing $D(\gamma(g)^{-1})$ commutes).

Compositions of commuting diagrams also commute, and 4.5 proves the statement. □

Lemma 4.7. *Let* $\mathbf{C}$ *be a model category, and* $F : \mathbf{C} \to \mathbf{D}$ *be a functor taking weak equivalences in* $\mathbf{C}$ *into isomorphisms in* $\mathbf{D}$*. If* $f \overset{l}{\sim} g$ *or* $f \overset{r}{\sim} g$*, then* $F(f) = F(g)$ *in* $\mathbf{D}$*.*

Proof. Intuitively, this special property of F ensures it behaves well on good cylinder objects, since $A \wedge I \sim A$.

$$A \coprod A \overset{i_0+i_1}{\hookrightarrow} A \wedge I \overset{w}{\to} A$$

$wi_0 = wi_1 (= \mathrm{id}_A) \implies F(w)F(i_0) = F(w)F(i_1) \overset{w \text{ weak equivalence}}{\implies} F(i_0) = F(i_1) \implies F(Hi_0) = F(Hi_1)$ for good left homotopy between f and g. □

This lemma has already indicated that the localisation of a model category $\mathbf{C}$ with respect to weak equivalences will preserve homotopies in $\mathbf{C}$, whereas the "smallest" such category might be the homotopy category. It also helps define maps from $\pi(A, X)$:

Proposition 4.8. *A cofibrant, X fibrant, then $\gamma : \mathrm{Hom}_{\mathbf{C}}(A, X) \to \mathrm{Hom}_{\mathrm{Ho}(\mathbf{C})}(A, X)$ is already surjective and induces a bijection $\pi(A, X) \cong \mathrm{Hom}_{\mathrm{Ho}(\mathbf{C})}(A, X)$.*

Proof. 4.7 ensures the induced $\gamma : \pi(A, X) \to \mathrm{Hom}_{\mathrm{Ho}(\mathbf{C})}(A, X)$ is well-defined. Then:

$$\pi(A, X) \overset{\gamma}{\to} \mathrm{Hom}_{\mathrm{Ho}(\mathbf{C})}(A, X) \overset{\text{def}}{=} \pi(RA, QX) \overset{p_X \cdot - \cdot i_A}{\longrightarrow} \pi(A, X)$$

where $p_X \cdot - \cdot i_A$ is bijective due to 3.6. The overall composition is a bijection (the identity), and thus the induced γ is also bijective. □

4.3 Localisation of Categories

The homotopy category actually depends only on the class of weak equivalences, which implies that weak equivalences carry the fundamental homotopy-theoretic information, while the cofibrations, fibrations, and the axioms they satisfy serve mainly as tools for making various constructions.

Definition 4.9. Let $W \subset \mathbf{C}$ be a class of morphisms. A functor $F : \mathbf{C} \to \mathbf{D}$ is a *localisation of* $\mathbf{C}$ *with respect to* W if:

(i) $F(f)$ is an isomorphism for each $f \in W$,

(ii) if $G : \mathbf{C} \to \mathbf{D}'$ is a functor carrying elements of W into isomorphisms, there exists unique $G' : \mathbf{D} \to \mathbf{D}'$ such that $G'F = G$.

We denote $\mathbf{C} \to W^{-1}\mathbf{C}$ the localisation, unique up to isomorphism.

The following example actually demonstrates a typical way to construct localisations: first verify the property of sending weak equivalences to isomorphisms; for the universal property, we need to specify a map in $\mathbf{C}$ for any map in $\mathbf{D}$ (by composing it with weak equivalences), and then define G' in terms of G.

Example 4.10. Let W be the class of morphisms in $\mathbf{Ab}$ of $f : A \to B$ such that $\ker(f)$ and $\mathrm{coker}(f)$ are torsion groups. Then the category $\mathbf{D}$ with the same objects and $\mathrm{Hom}_{\mathbf{D}}(A, B) = \mathrm{Hom}_{\mathbf{Ab}}(\mathbb{Q} \otimes A, \mathbb{Q} \otimes B)$, with $F : \mathbf{Ab} \to \mathbf{D}$ sending f to $\mathrm{id} \otimes f$, is a localisation of $\mathbf{Ab}$ with respect to W.

The weak equivalences are sent to isomorphisms: $\ker(\mathrm{id} \otimes f) \cong \mathbb{Q} \otimes \ker(f) = 0$ since $q \otimes k = \frac{q}{n} \otimes nk = 0$, and similar for $\mathrm{coker}(f)$.

There is a weak equivalence $i_A : A \to \mathbb{Q}\otimes A$, and we can write any $u \in \mathrm{Hom}_{\mathbf{Ab}}(\mathbb{Q}\otimes A, \mathbb{Q}\otimes B)$ as $u = F(i_B)^{-1}F(u)F(i_A)$ for $u : A \to B$ since the functor onto $\mathrm{Hom}_{\mathbf{D}}(\mathbb{Q}\otimes A, \mathbb{Q}\otimes B)$ is surjective (similar to what we have seen!). Then define $G'(u) = G(i_B)^{-1}G(u)G(i_A)$, and by the commutative diagram this is the G' we desired.

Theorem 4.11. *Let $\mathbf{C}$ be a model category and $W \subset \mathbf{C}$ the class of weak equivalences. Then the functor $\gamma : \mathbf{C} \to \mathrm{Ho}(\mathbf{C})$ is a localisation of $\mathbf{C}$ with respect to W.*

Proof. 4.5 directly gives the first condition. Suppose that we are given a functor $G : \mathbf{C} \to \mathbf{D}$, and we're constructing a unique $G' : \mathrm{Ho}(\mathbf{C}) \to \mathbf{D}$ such that $G = G'\gamma$. Given $f \in \mathrm{Hom}_{\mathrm{Ho}(\mathbf{C})}(X, Y)$, write $f = [f']$, where $f' \in \mathrm{Hom}_{\mathbf{C}}(RQX, RQY)$.

Uniqueness: by 4.5, $f = \gamma(p_Y)\gamma(i_{QY})\gamma(f')\gamma(i_{QX})\gamma(p_X)^{-1}$, and G' has to be of the form:

$$G'(f) = G(p_Y)G(i_{QY})^{-1}G(f')G(i_{QX})G(p_X)^{-1}$$

. Then we define G' in this way. G' is clearly a functor as it preserves the identity and compositions.

Existence: if $\gamma(h) = f$, then by definition of γ, after shifting f' up to homotopy, we have a commutative diagram:

$$\begin{array}{ccccc} X & \xleftarrow[p_X]{} & QX & \xrightarrow{i_{QX}} & RQ(X) \\ \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle \tilde{h}} & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p_Y]{} & QY & \xrightarrow{i_{QY}} & RQ(Y) \end{array}$$

Applying G to the diagram and using the "weak equivalence to isomorphism" property, we have $G'(f) = G(h)$: specifically, $G(f')G(i_{QX}) = G(i_{QY})G(\tilde{h})$ gives $G(\tilde{h}) = G(i_{QY})^{-1}G(f')G(i_{QX})$ due to the property, and similarly we can express $G(h)$ in terms of the compositions, which is $G'(f)$. Then $G\gamma = G'$.

□

5 Derived functors and Quillen equivalences

5.1 Quillen adjunction

In this section, we clarify the concept of morphisms between model categories, which we call Quillen adjunctions:

Definition 5.1. Let $\mathbf{C}$ and $\mathbf{D}$ be model categories.

1. A functor $F : \mathbf{C} \to \mathbf{D}$ is a *left Quillen functor* if F is a left adjoint and preserves cofibrations and trivial cofibrations.
2. A functor $U : \mathbf{C} \to \mathbf{D}$ is a *right Quillen functor* if U is a right adjoint and preserves fibrations and trivial fibrations.
3. We call (F, U, ϕ) a Quillen adjunction if F is a left Quillen functor, where ϕ is the natural isomorphism of adjoints.

The next lemma implies that left Quillen functors actually preserve weak equivalences between cofibrant objects.

Lemma 5.2 (Ken Brown's lemma)**.** *Suppose $\mathbf{C}, \mathbf{D}$ are model categories. If F is a functor taking trivial cofibrations between cofibrant objects to weak equivalences, then F takes all weak equivalences between cofibrant objects to weak equivalences. The dual statement also holds.*

Proof. Let $f : A \to B$ be a weak equivalence between two cofibrant objects. Proving $F(f)$ to be a weak equivalence also uses one side of the coproduct.

Let $(f, 1_B) : A \coprod B \to B$, and factorise $(f, 1_B) = pq$ for cofibration q and trivial fibration p. Using the pushout diagram

$$\begin{array}{ccc} 0 & \longrightarrow & A \\ \downarrow & & \downarrow{\scriptstyle i_1} \\ B & \xrightarrow{i_2} & A \coprod B \end{array}$$

we see i_1 and i_2 are both cofibrations. Note that $f, 1_B$ and p being weak equivalences implies $q \circ i_1$ and $q \circ i_2$ are weak equivalences. By hypothesis $F(q \circ i_1)$ and $F(q \circ i_2)$ are weak equivalences, and $F(p \circ q \circ i_2) = F(1_B)$ is a weak equivalence $\implies$ $F(p)$ is a weak equivalence $\implies$ $F(f) = F(p) \circ F(q \circ i_1)$ is a weak equivalence.

□

The next lemma provides a short route in finding and verifying Quillen adjunctions:

Lemma 5.3. *Let (F, U, ϕ) be adjunctions between model categories $\mathbf{C}$ and $\mathbf{D}$. Then it is a Quillen adjunction if and only if U (resp. F) is a right (resp. left) Quillen functor.*

Proof. The proof basically follows from the naturality of ϕ. The naturality and adjointness imply that Ff has the left lifting property with respect to g if and only if f has the left lifting property with respect to Ug. All (trivial) cofibrations are characterised by lifting properties: g is fibration $\implies$ Ff has LLP with respect to g for f trivial cofibration $\implies$ f has LLP with respect to Ug $\implies$ Ug is a fibration. □

Example 5.4. For a model category $\mathbf{C}$, let $\mathbf{C}_*$ be the category of (X, v), where there is a map $* \xrightarrow{v} X$. A morphism from (X, v) to (Y, w) is $X \to Y$ such that the diagram commutes, and the weak equivalences, fibrations, and cofibrations are induced by those of $X \to Y$. We have an obvious functor from $\mathbf{C}$ to $\mathbf{C}_*$ taking X to $X \coprod *$, which is left adjoint to the forgetful functor U. Clearly by definition U is a right Quillen functor, and by the previous lemma we know F and U form a Quillen adjunction.

If (F, G, ϕ) and (F', G', ϕ') are two pairs of adjunctions, then clearly we can compose them as $(F' \circ F, G' \circ G, \phi' \circ \phi)$, and compositions of Quillen adjunctions are still Quillen adjunctions.

5.2 Derived functors

A Quillen adjunction between model categories also induces an adjunction between their homotopy categories, which we define below.

Definition 5.5. Let $\mathbf{C}$ and $\mathbf{D}$ be model categories, and let F and G be left and right Quillen functors. Let Q be the cofibrant replacement in $\mathbf{C}$ as defined in 4.2, and let R be the fibrant replacement in $\mathbf{D}$. By 4.11 and the definition of localisation, if $\psi : \mathbf{C} \to \mathbf{D}$ is a functor taking weak equivalences to weak equivalences, then it induces a functor $\mathrm{Ho}(\psi) : \mathrm{Ho}(\mathbf{C}) \to \mathrm{Ho}(\mathbf{D})$ since the composition $\mathbf{C} \to \mathbf{D} \to \mathrm{Ho}(\mathbf{D})$ takes weak equivalences to isomorphisms.

1. Since by 4.2 $Q : \mathbf{C} \to \mathbf{C}_c$ preserves weak equivalences and by 5.2 $F : \mathbf{C}_c \to \mathbf{D}$ also preserves weak equivalences, they induce functors $\mathrm{Ho}(Q)$ and $\mathrm{Ho}(F)$. We define the *total left derived functor* of F, called LF, to be the composition
$$\mathrm{Ho}(\mathbf{C}) \xrightarrow{\mathrm{Ho}Q} \mathrm{Ho}(\mathbf{C}_c) \xrightarrow{\mathrm{Ho}F} \mathrm{Ho}(\mathbf{D}).$$
2. Dually, we define the *total right derived functor* of G, called RG, to be the composition
$$\mathrm{Ho}(\mathbf{D}) \xrightarrow{\mathrm{Ho}R} \mathrm{Ho}(\mathbf{D}_f) \xrightarrow{\mathrm{Ho}G} \mathrm{Ho}(\mathbf{C}).$$

One example of derived functors would be homotopy limits and homotopy colimits, which we will explain later. Another classical example is the derived tensor and derived Hom in a monoidal model category (see [24, Chapter 4] for more details). In a monoidal model category $\mathbf{C}$, $\otimes$ and Hom are intrinsically Quillen bifunctors and form a Quillen adjunction, which induces a derived adjunction $\otimes^L$ and $\mathbb{R}\mathrm{Hom}$. The category of chain complexes over a ring R, $\mathbf{Ch}(R)$ is a monoidal model category with its projective model structure (which we will explain later), and for R-module M and N viewed as chain complexes at degree 0, one recognises that $H^n(\mathbb{R}\mathrm{Hom}_R(M, N))$ is precisely $\mathrm{Ext}^n_R(M, N)$ by definition of a right derived functor.

The operation of taking the total derived functor is almost functorial, in the sense of the following theorem.

Theorem 5.6. *For any model category $\mathbf{C}$, there is a natural isomorphism $\alpha : L(1_{\mathbf{C}}) \to 1_{\mathrm{Ho}(\mathbf{C})}$ and a natural isomorphism $m : LF' \circ LF \to L(F' \circ F)$. The same holds for right derived functors.*

Proof. For each X, we define α_X to be $\mathrm{Ho}(q_X)$, where $q_X : QX \to X$ is the natural trivial fibration from QX to X. The following diagram commutes:

$$\begin{array}{ccc} QX & \xrightarrow{\mathrm{Ho}(q_X)} & X \\ \downarrow{\scriptstyle Q(f)} & & \downarrow{\scriptstyle f} \\ QY & \xrightarrow{\mathrm{Ho}(q_Y)} & Y \end{array}$$

Clearly each q is a weak equivalence and becomes an isomorphism on $\mathrm{Ho}(\mathbf{C})$.

We define m_X to be the natural map

$$m_X : (LF')(LF) = F'QFQX \xrightarrow{F'q_{FQX}} F'FQX = L(F'F)X.$$

Since both the source and target of m preserves weak equivalences, m is indeed a natural transformation on $\mathrm{Ho}(\mathbf{C})$. Each q_{FQX} is a weak equivalence and is preserved by F', and thus m is a natural isomorphism.

□

The next lemma clarifies our use of the word "derived adjunction".

Lemma 5.7. *Let $\mathbf{C}$ and $\mathbf{D}$ be model categories and (F, U, ϕ) be a Quillen adjunction. Then there is an adjunction $L(F, U, \phi) := (LF, RU, R\phi)$ which we call the derived adjunction.*

Proof. The key is to show $R\phi$ respects homotopy relations. See [Hov, Lem 1.3.10]. □

5.3 Quillen equivalence

Definition 5.8. A Quillen adjunction (F, U, ϕ) is a *Quillen equivalence* if for all cofibrant X in $\mathbf{C}$ and fibrant Y in $\mathbf{D}$, a map $f : FX \to Y$ is a weak equivalence in $\mathbf{D}$ if and only if $\phi(f) : X \to UY$ is a weak equivalence in $\mathbf{C}$.

A key property of Quillen equivalence is that they induce an equivalence of homotopy categories.

Proposition 5.9. *Let $(F, U, \phi) : \mathbf{C} \to \mathbf{D}$ be a Quillen adjunction. The following are equivalent:*

1. *(F, U, ϕ) is a Quillen equivalence.*
2. *The composition $X \xrightarrow{\eta} UFX \xrightarrow{Ur_{FX}} URFX$ is a weak equivalence for all cofibrant X, and $FQUX \xrightarrow{FqUX} FQX \xrightarrow{\epsilon} X$ for all fibrant X.*
3. *$L(F, U, \phi)$ is an adjoint equivalence of categories.*

Proof. See [24, Proposition 1.3.13]. □

This induces an even more useful criterion formulated in the language of weak equivalences.

Corollary 5.10. *Let $(F, U, \phi) : \mathbf{C} \to \mathbf{D}$ be a Quillen adjunction. The following are equivalent:*

1. *(F, U, ϕ) is a Quillen equivalence.*
2. *If f is a map between cofibrant objects, then f is a weak equivalence whenever Ff is; for every fibrant Y, the map $FQUY \to Y$ is a weak equivalence.*
3. *If g is a map between fibrant objects, then g is a weak equivalence whenever Ug is; for every cofibrant X, the map $X \to URFX$ is a weak equivalence.*

Proof. See [24, Proposition 1.3.16]. □

We will come across some examples of Quillen equivalences. A straightforward example is the identity equivalence between projective model structure of $\mathbf{Ch}(R)$ and the injective model structure of $\mathbf{Ch}(R)$. Another important example is the geometric realisation.

6 Cofibrantly generated model categories

6.1 Ordinals and cardinals

We need to recall some basic facts about ordinals and cardinals.

Definition 6.1 (Ordinals)**.** An ordinal is a transitive set well-ordered by $\in$.

Remark. A set α is transitive if $x \in y \in \alpha \implies x \in \alpha$; equivalently, every element of α is also a subset of α, straight from the definition. For example, $2 = \{0, 1\} = \{\emptyset, \{\emptyset\}\}$ and $\omega = \{0, 1, 2, \dots\}$ are ordinals. We can order ordinals by $\alpha < \beta$ if $\alpha \in \beta$ so that we may talk about "smaller" or "greater" ordinals; in this way, an ordinal is a well-ordered set of smaller ordinals, which we often think of as a poset category, and ω is said to be the smallest infinite ordinal.

Definition 6.2 (Cardinality, cardinals, successor ordinal, and limit ordinal)**.** Given any set A, the cardinality $|A|$ of A is the smallest ordinal such that there is a bijection $|A| \to A$. A cardinal is an ordinal κ such that $\kappa = |\kappa|$.

A successor ordinal is an ordinal α that can be written as $\alpha = \beta \cup \{\beta\}$ for some ordinal β. A limit ordinal is an ordinal that is neither 0 nor a successor ordinal, e.g. ω.

Equivalently, a cardinal is an ordinal that has greater cardinality than any smaller ordinal. The following definition is widely used in the context of locally presentable categories.

Definition 6.3. A cardinal κ is *regular* if for every subset $S \subset \kappa$ such that $|S| < \kappa$, we have $\sup S < \kappa$.

Here comes the most important definition.

Definition 6.4. Let κ be a cardinal. An ordinal α is κ-filtered if it is a limit ordinal and, if $A \leq \alpha$ and $|A| \leq \kappa$, then $\sup A < \alpha$.

Remark. Intuitively, this says α cannot be approached by any set of size at most κ. Equivalently, α is κ-filtered if and only if the cofinality of α (minimum number of steps to approach α) is larger than κ.

From the definition, a regular cardinal κ is λ-filtered as a limit ordinal for every cardinal $\lambda < \kappa$. Recall that $\aleph_1, \aleph_2, \dots$ are successor cardinals after the first infinite cardinal $\aleph_0 = |\omega|$. $\aleph_0$ is regular since the supremum of finitely many natural numbers is still a natural number, and it turns out $\aleph_1, \aleph_2, \dots$ are also regular. However, $\aleph_\omega = \sup_{n<\omega} \aleph_n$ is not regular since it is the supremum of only countably many smaller cardinals, whose cardinality is less than or equal to $\aleph_0$.

Example 6.5. Every successor cardinal γ^+ larger than the infinite cardinal γ is γ-filtered. For example, $\aleph_1$ is $\aleph_0$-filtered, but $\aleph_\omega$ is not, since the cofinal subset of $\aleph_\omega$, $\{\aleph_n : n \leq \omega\}$, has cardinality ω. There are certainly other γ-filtered ordinals, such as $2 \cdot \gamma_1$.

6.2 Small objects

Definition 6.6. Let $\mathbf{C}$ be a model category with all colimits, and let λ be an ordinal. A λ-sequence in $\mathbf{C}$ is a colimit preserving functor $X : \lambda \to \mathbf{C}$,

$$X_0 \to X_1 \to \cdots \to X_\beta \to \dots$$

Preserving colimits gives $X_\gamma = \operatorname{colim}_{\beta<\gamma} X_\beta$ for all limit ordinals $\gamma < \lambda$. The map $X_0 \to \operatorname{colim}_{\beta<\lambda} X_\beta$ is the (transfinite) *composition* of the λ-sequence, and if every map $X_\beta \to X_{\beta+1}$ is in a collection of morphisms D, then it is called the *transfinite composition* of maps of D.

Below is the key definition. A small object is essentially an object such that for some cardinal κ, its map into any target beyond that cardinal can be built from previous steps.

Definition 6.7. Suppose D is a collection of morphisms in a model category $\mathbf{C}$, A is an object of $\mathbf{C}$ and κ is a cardinal. A is *κ-small relative to D* if, for any κ-filtered ordinals λ and all λ-sequences

$$X_0 \to X_1 \to \cdots \to X_\beta \to \dots$$

such that each map $X_\beta \to X_{\beta+1}$ is in D for $\beta + 1 < \lambda$, the map of sets

$$\mathrm{colim}_{\beta<\lambda} \mathrm{Hom}_{\mathbf{C}}(A, X_\beta) \to \mathrm{Hom}_{\mathbf{C}}(A, \mathrm{colim}_{\beta<\lambda} X_\beta)$$

is an isomorphism. We say A is *small relative to D* if it is κ-small relative to D for some κ. We say A is *small* if it is small relative to all morphisms in $\mathbf{C}$.

Example 6.8. Every set is small. Let $\mathbf{C} = \mathbf{Sets}$. For any set A, we claim A is $|A|$-small. Given a $|A|$-filtered ordinal λ and a λ-sequence X, we prove it by showing the map

$$\mathrm{colim}_{\beta<\lambda} \mathrm{Hom}_{\mathbf{C}}(A, X_\beta) \to \mathrm{Hom}_{\mathbf{C}}(A, \mathrm{colim}_{\beta<\lambda} X_\beta)$$

is injective and surjective. Surjectivity is equivalent to finding a map $g \in \mathrm{Hom}_{\mathbf{C}}(A, X_{\beta_a})$ as a factor of every $f \in \mathrm{Hom}_{\mathbf{C}}(A, \mathrm{colim}_{\beta<\lambda} X_\beta)$, for some X_{β_a}. In the category of **Sets**, $\mathrm{colim}_{\beta<\lambda} X_\beta$ is a disjoint union identifying each element x with its image in latter stages (then the images of X_β form nested sets). Therefore, give $f \in \mathrm{Hom}_{\mathbf{C}}(A, \mathrm{colim}_{\beta<\lambda} X_\beta)$, for every $a \in A$ there exists β_a such that $f(a)$ is in the image of X_{β_a}. Let $\gamma = \sup_{a\in A} \beta_a$. By definition of $|A|$-filtered λ, we have $\gamma < \lambda$, and thus f factors through some map $g \in \mathrm{Hom}_{\mathbf{C}}(A, X_\gamma)$. This shows surjectivity.

Injectivity is similar: for $f = f'$ in the limit stage, for each a we must have $f(a) = f'(a)$ in the image of X_{β_a}, and $|A|$-filtration implies $f = f'$ at some stage of (image of) X_γ.

Example 6.9. Every R-module for ring R is also small. Let $\mathbf{C} = R$-**mod** and $\kappa = |A|(|A| + |R|)$. For a κ-filtered ordinal λ and a λ-sequence X, we can still factor every $f \in \mathrm{Hom}_{\mathbf{C}}(A, \mathrm{colim}_{\beta<\lambda} X_\beta)$ into some $g \in \mathrm{Hom}_{\mathbf{Sets}}(A, X_{\beta_a})$. This is because the argument above does not rely on the actual category (that the images of X_β are nested follow directly from the definition of colimits). Therefore,

$$\mathrm{colim}_{\beta<\lambda} \mathrm{Hom}_{\mathbf{C}}(A, X_\beta) \to \mathrm{Hom}_{\mathbf{C}}(A, \mathrm{colim}_{\beta<\lambda} X_\beta)$$

is also injective, and it remains to produce an R-linear map out of g. We know that for any pair $(x, y) \in A \times A$, $g(x + y) = g(x) + g(y)$ in $\mathrm{colim}_{\beta<\lambda} X_\beta$, and by the above argument, they must equal in some earlier stage $X_{\beta_{(x,y)}}$. Similarly, $g(rx) = rg(x)$ in some $X_{\beta_{(r,x)}}$. The set of all such objects has cardinal less than κ, and thus their supremum $\gamma < \lambda$ as above. Thus f also factors through an R-linear map g', which implies every R-module is small.

6.3 The small object argument

Definition 6.10. Let I be a class of map in a category $\mathbf{C}$.

1. A map is *I-injective* if it has the right lifting property with respect to every map in I. The class of I-injective maps is denoted by I-inj,
2. A map is an *I-cofibration* if it has the left lifting property with respect to every I-injective map. The class of I-cofibrations is denoted by I-cof,

Definition 6.11. Let I be a class of maps in a category $\mathbf{C}$. A *relative I-cell complex* is a transfinite composition of pushouts of elements of I. The class of relative I-cells is denoted by I-cell.

Namely, if $f : A \to B$ is in I-cell, then there is an ordinal λ and a λ-sequence X such that $X_0 = A$ and $X_\lambda = \mathrm{colim}_{\beta<\lambda} X_\beta = B$, and for each $\beta + 1 < \lambda$, there is a pushout square

$$\begin{array}{ccc} C_\beta & \longrightarrow & X_\beta \\ \downarrow{\scriptstyle h_\beta} & & \downarrow \\ D_\beta & \longrightarrow & X_{\beta+1} \end{array}$$

such that g_β is in I. We say an object A in $\mathbf{C}$ is an *I-cell complex* if $0 \to A$ is a relative I-cell complex.

Lemma 6.12. *I-cell $\subset$ I-cof.*

Proof. Since $I \subset I$-cof, it suffices to show that I-cof is closed under pushouts and (transfinite) compositions. Since I-cofibrations are defined by lifting properties, this is clear from the diagrams. Assume f has the left lifting property with respect to h and g is the pushout of f:

$$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & E \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & \nearrow{\scriptstyle g'} & \downarrow{\scriptstyle h} \\ C & \longrightarrow & D & \longrightarrow & F \end{array}$$

where left lifting property of f gives $C \to E$, and pushouts give g'. If now f and g have left lifting property with h:

$$\begin{array}{ccc} A & \longrightarrow & D \\ \downarrow{\scriptstyle f} & \nearrow{\scriptstyle g'} & \\ B & \overset{f'}{\nearrow} & \downarrow{\scriptstyle h} \\ \downarrow{\scriptstyle g} & & \\ C & \longrightarrow & E \end{array}$$

where f' is given by the left lifting property of f and g' is then given by g. Thus I-cof is closed under pushouts and compositions.

We have already proved the claim at the successor stage of the transfinite composition (which is the same as proving for two compositions), and it remains to show for the limit stage. Assume γ is a limit ordinal, and by definition we have $X_\gamma = \mathrm{colim}_{\beta<\gamma} X_\beta$. By induction hypothesis, we assume that $f_\beta : X_0 \to X_\beta$ has the left lifting property with respect to h for any $\beta < \gamma$, where h is any map in some collection of morphisms. Suppose we have the following diagram:

$$\begin{array}{ccc} X_0 & \longrightarrow & D \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle h} \\ X_\gamma & \longrightarrow & E \end{array}$$

Then this induces a similar diagram for f_β by the colimit structure maps, so we obtain $g_\beta : X_\beta \to D$ for any $\beta < \gamma$, which are compatible with the commutative diagrams at previous

stages (which can be proven during the induction as part of the induction hypothesis). Then by definition of colimit we obtain $g : X_\lambda \to D$, the desired lift.

Thus the transfinite composition $X_0 \to X_\lambda$ also have the left lifting property with any h by the transfinite induction.

□

Lemma 6.13. *Suppose $X : \lambda \to \mathbf{C}$ is a λ-sequence such that each map $X_\beta \to X_{\beta+1}$ is either a pushout or an isomorphism. Then the transfinite composition of X is a relative I-cell complex.*

Proof. The idea is to define an equivalence relation on the ordinal λ for $\alpha \leq \beta$: if $\alpha \leq \beta$, define $\alpha \sim \beta$ if the map $X_\gamma \to X_{\gamma+1}$ is an isomorphism for all $\alpha \leq \gamma < \beta$. The set of these equivalence classes clearly forms a well-ordered set and is isomorphic to a unique ordinal μ. Then this induces a functor $Y : \mu \to \mathbf{C}$, whose transfinite composition is naturally isomorphic to that of X.

□

The following property of I-cell is crucial to the small object argument.

Lemma 6.14. *I-cell is closed under transfinite compositions, and any pushout of coproducts of maps in I is also in I-cell.*

Proof. Let $X : \lambda \to \mathbf{C}$ be a λ-sequence such that each $X_\beta \to X_{\beta+1}$ is a relative cell complex with respect to a γ_β sequence. Then there is a natural total order on the set of $\{(\beta, \gamma) : \beta < \lambda, \gamma < \sup \gamma_\beta\}$, which is then isomorphic to some ordinal μ. Then there is a functor $Z : \mu \to \mathbf{C}$ such that each map is either a pushout (maps within each $Y_\beta : \gamma_\beta \to \mathbf{C}$) or an isomorphism (the map connecting β and $\beta + 1$). By the previous lemma the composition of Z is in I-cell.

Now suppose $g_k : C_k \to D_k$ is a map in I for $k \in K$. Then we can build transfinitely the following map f:

$$\begin{array}{ccc} \coprod C_k & \longrightarrow & X \\ \downarrow{\scriptstyle \coprod g_k} & & \downarrow{\scriptstyle f} \\ \coprod D_k & \longrightarrow & Y \end{array}$$

Essentially, pushout of coproducts can be built from pushout of individual elements. We construct $X : \lambda \to \mathbf{C}$ as follows, where λ is an ordinal isomorphic to K. $X_0 = X$, and at each stage β, there is a pushout:

$$\begin{array}{ccc} C_\beta & \longrightarrow & X_\beta \\ \downarrow{\scriptstyle g_\beta} & & \downarrow \\ D_\beta & \longrightarrow & X_{\beta+1} \end{array}$$

and at limit ordinal κ, $X_\kappa = \operatorname{colim}_{\beta<\kappa} X_\beta$. This colimit is isomorphic to Y since $\operatorname{colim}_{\beta<\lambda} X_\beta$ also satisfies the universal property of coproducts. Given any Z and a commutative diagram of

coproducts, we naturally have a commutative diagram as follows when $\beta = 0$:

$$\begin{array}{ccccc} C_\beta & \longrightarrow & \coprod C_k & \longrightarrow & X_\beta \\ \downarrow{\scriptstyle g_\beta} & & \downarrow{\scriptstyle \coprod g_k} & & \downarrow \\ D_\beta & \longrightarrow & \coprod D_k & \longrightarrow & Z \end{array}$$

Then, since X_1 is built as a pushout, we can factor $X \to Y$ into $X_1 \to Y$ and land in the case when $\beta = 1$ in the above. Inductively, we obtain $X_\beta \to Z$ for every β, and thus there is a map $\mathrm{colim}_{\beta<\lambda} X_\beta \to Z$ satisfying the pushout diagram.

□

Theorem 6.15 (The small object argument)**.** *Let* $\mathbf{C}$ *be a cocomplete category, and* I *a set of morphisms in* $\mathbf{C}$*. Suppose the domains of* I *are small relative to* I*-cell. Then there is a functorial factorisation* (γ, δ) *such that* $f = \delta(f)\gamma(f)$*, where* $\gamma(f)$ *is in* I*-cell and* $\delta(f)$ *is in* I*-inj.*

Proof. First choose an cardinal κ such that the domain of every map in I is κ-small relative to I-cell, and let λ be a κ-filtered ordinal. The basic idea is to factorise f as a transfinite composition of some functorial λ-sequence Z^f, which we call $\gamma(f)$, and a map $\delta(f) : E_f := \mathrm{colim}_{\beta<\lambda} Z^f_\beta \to Y$ induced by collection of map $\rho^f_\beta : Z^f_\beta \to Y$ which also factorises f.

Before the formal proof, we will give a brief idea of the reasons behind our proof.

The key is to prove $\gamma(f)$ to be in I-cell. To do this, we need to show it has the right lifting property with some map in I:

$$\begin{array}{ccc} A & \xrightarrow[h]{} & E_f \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle \delta(f)} \\ B & \xrightarrow{k} & Y \end{array} \tag{1}$$

where g is a map in I. Smallness comes in here: since the domains of I are κ-small relative to I-cell, and we will construct Z^f to be a λ-sequence with maps in I-cell, we see h factorises as $A \xrightarrow{h_\beta} Z^f_\beta \to E_f$ at some previous stage β. To obtain a map $B \to E_f$, one way is to specify for every β and every such A, B, g a pushout square factorising the square

$$\begin{array}{ccc} A & \xrightarrow[h_\beta]{} & Z^f_\beta \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle \rho^f_\beta} \\ B & \xrightarrow{k} & Y \end{array}$$

A natural thought is something close to a pushout

$$\begin{array}{ccc} A & \xrightarrow[h_\beta]{} & Z^f_\beta \\ \downarrow{\scriptstyle g} & & \downarrow \\ B & \longrightarrow & Z^f_{\beta+1} \end{array}$$

so that $B \to Z^f_{\beta+1} \to E_f$ is a good choice satisfying all the good properties; it turns out this pushout is constructed using coproducts of all such A and B.

Note: pushouts are defined by construction and used to formulate the desired lift $B \to E_f$ (also ensuring the definition of I-cell), while the commutative squares that we verify along the proof serve to propel the induction. With this in mind, we are ready to build Z^f and ρ^f transfinitely. The rest is the formal proof.

First, for every successor ordinal $\beta + 1$, suppose we have defined Z^f_β and ρ^f_β. Let S be the indexing set of all commutative squares

$$\begin{array}{ccc} A & \longrightarrow & Z^f_\beta \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle \rho^f_\beta} \\ B & \longrightarrow & Y \end{array} \tag{2}$$

where g is a map in I. For $s \in S$, denote $g_s : A_s \to B_s$ as the corresponding map in I. Then define $Z^f_{\beta+1}$ to be the pushout of the diagram:

$$\begin{array}{ccc} \coprod_{s\in S} A_s & \longrightarrow & Z^f_\beta \\ \downarrow{\scriptstyle \coprod_{s\in S} g_s} & & \downarrow \\ \coprod_{s\in S} B_s & \longrightarrow & Z^f_{\beta+1} \end{array}$$

where from 6.14, we know the map $Z^f_\beta \to Z^f_{\beta+1}$ is also in I-cell. Properties of coproducts ensures the existence of $\coprod_{s\in S} A_s \to Z^f_\beta$, and this pushout property can actually be interpreted individually. Consider the following diagram induced by 2:

$$\begin{array}{ccccc} A & \longrightarrow & \coprod_{s\in S} A_s & \longrightarrow & Z^f_\beta \\ & & \downarrow & \searrow^{\coprod_{s\in S} g_s} & \downarrow \\ & & B & \longrightarrow & \coprod_{s\in S} B_s \longrightarrow Y \end{array}$$

Thus for any diagram like 2, we indeed obtain a map $B \to Z^f_{\beta+1}$ satisfying good properties since $Z^f_{\beta+1}$ is the pushout of coproducts. $\rho^f_{\beta+1} : Z^f_{\beta+1} \to Y$ is also induced in this way.

Then for limit ordinals γ, define $Z^f_\gamma = \operatorname{colim}_{\beta<\gamma} Z^f_\beta$, a natural choice for colimit-preserving Z^f. The transfinite construction is functorial: coproducts and pushouts of maps are functorial. $\gamma(f)$ as the transfinite composition of the pushouts of coproducts of maps in I is in I-cell. For $\delta(f)$ in 1, previous arguments have shown that smallness of A gives

$$\begin{array}{ccc} A & \xrightarrow[h_\beta]{} & Z^f_\beta \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle \rho^f_\beta} \\ B & \xrightarrow{k} & Y \end{array}$$

where ρ^f_β is $Z^f_\beta \to E_f \overset{\delta(f)}{\to} Y$. By construction above, there is a map $B \overset{d}{\to} Z^f_{\beta+1}$ such that $dg = ih_\beta$ and $k = \rho^f_{\beta+1} d$, where $i : Z^f_\beta \to Z^f_{\beta+1}$. Then $B \overset{d}{\to} Z^f_{\beta+1} \to E_f$ is the required lift in 1. Hence $\delta(f)$ is in I-inj, and the theorem is proved.

□

The following two corollaries lie under the important properties of cofibrantly generated model categories that we will discuss next section.

Corollary 6.16. *Suppose the domains of maps are small relative to I-cell. If $f : A \to B$ is in I-cof, it is the retract of a map $g : A \to C$ by a map that fixes A.*

Proof. This immediately follows from the retract argument in 2.7. By 6.15, $f = pg$ where p is in I-inj, and thus f has left lifting property with respect to $p : C \to B$. There is a lift $i : B \to C$, which forms the retract diagram:

$$\begin{array}{ccccc} A & \xrightarrow{\text{id}} & A & \xrightarrow{\text{id}} & A \\ {\scriptstyle f}\downarrow & & {\scriptstyle g}\downarrow & & {\scriptstyle f}\downarrow \\ B & \xrightarrow[i]{} & C & \xrightarrow{p} & B \end{array}$$

□

Corollary 6.17 (Hirschhorn)**.** *Let I be a set of maps in a cocomplete category $\mathbf{C}$. Suppose the domains of I are small relative to I-cell, and A is an object small relative to I-cell. Then A is in fact small relative to I-cof.*

Proof. Suppose A is κ-small relative to I-cell. Let λ be a κ-filtered ordinal, and let $X : \lambda \to \mathbf{C}$ be a λ-sequence of I-cofibrations f_β. The idea is to construct a λ-sequence Y of relative I-cell complexes g_β such that maps of X are retracts of those of Y. It relies on a retract argument similar to above.

Let $Y_0 = X_0$, $r_0 = i_0 = \text{id}$. We will construct Y_β, $i_\beta : X_\beta \to Y_\beta$, $r_\beta : Y_\beta \to X_\beta$ such that $r_\beta i_\beta = \text{id}$ transfinitely inductively. With Y_β, r_β, i_β defined, we factorise $Y_\beta \to X_\beta \to X_{\beta+1}$ by 6.15 as $Y_\beta \overset{g_\beta}{\to} Y_{\beta+1} \overset{r_{\beta+1}}{\to} X_{\beta+1}$, where g_β is in I-cell and $r_{\beta+1}$ is in I-inj. Then we have the following diagram:

$$\begin{array}{ccc} X_\beta & \xrightarrow{g_\beta i_\beta} & Y_{\beta+1} \\ {\scriptstyle f_\beta}\downarrow & \overset{\exists i_{\beta+1}}{\nearrow} & {\scriptstyle r_{\beta+1}}\downarrow \\ X_{\beta+1} & \xrightarrow{\text{id}} & X_{\beta+1} \end{array}$$

where the diagram commutes since $r_{\beta+1} g_\beta i_\beta = f_\beta r_\beta i_\beta = f_\beta$. By left lifting property of f_β, there is a lift $i_{\beta+1} : X_{\beta+1} \to Y_{\beta+1}$. For limit ordinals γ, define Y_γ, i_γ, and r_γ to be the natural colimits.

Then the maps i_β and r_β canonically induce the following diagram:

$$\begin{array}{ccccc} \operatorname{colim}_\beta \operatorname{Hom}(A, X_\beta) & \xrightarrow{i} & \operatorname{colim}_\beta \operatorname{Hom}(A, Y_\beta) & \xrightarrow{r} & \operatorname{colim}_\beta \operatorname{Hom}(A, X_\beta) \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} \\ \operatorname{Hom}(A, \operatorname{colim}_\beta X_\beta) & \xrightarrow[i]{} & \operatorname{Hom}(A, \operatorname{colim}_\beta Y_\beta) & \xrightarrow[r]{} & \operatorname{Hom}(A, \operatorname{colim}_\beta X_\beta) \end{array}$$

where i is induced by i_β and r is induced by r_β. By assumption g is an isomorphism. From $gi = if$ we have f is injective, and from $fr = rg$ we know f is surjective. Hence f is also an isomorphism, and A is in fact small relatve to I-cof.

□

6.4 Cofibrantly generated model categories

Let $\mathbf{C}$ be a model category. $\mathbf{C}$ is cofibrantly generated if there is a set of *generating cofibrations* I and a set of *generating trivial cofibrations* J such that:

1. The domains of I are small relative to I-cell, and the domains of J are small relative to J-cell;
2. The class of fibrations is J-inj, and the class of trivial fibrations is I-inj.

The following proposition summarises the basic properties of cofibrantly generated model categories.

Proposition 6.18. *Suppose $\mathbf{C}$ is a cofibrantly generated model category with generating cofibrations I and generating trivial cofibrations J. Then:*

1. *The cofibrations are I-cof, and the trivial cofibrations are J-cof.*
2. *Every cofibration is a retract of a relative I-cell complex, and every trivial cofibration is a retract of a relative J-cell complex.*
3. *The domains of I are small relative to the cofibrations, and the domains of J are small relative to the trivial cofibrations.*

Proof. 1. follows from the definition, 2. follows from 6.16, and 3. follows from 6.17.

□

Most model categories we will encounter are cofibrantly generated, and the following construction theorem provides a practical way to detect them. It is crucial to the examples afterwards and often used as a central theorem to prove a model category.

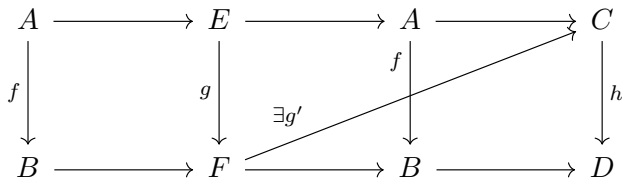
Theorem 6.19. *Suppose $\mathbf{D}$ is a subcategory of a complete and cocomplete category $\mathbf{C}$, and I, J are sets of maps in $\mathbf{C}$. Then there is a cofibrantly generated model category with I being the generating cofibrations, J being the generating trivial cofibrations, and $\mathbf{D}$ being the subcategory of weak equivalences if and only if the following are satisfied:*

1. *$\mathbf{D}$ has the two out of three property and is closed under retracts.*
2. *The domains of I are small relative to I-cell, and domains of J are small relative to J-cell.*
3. *J-cell $\subset \mathbf{D} \cap I$-cof, and I-inj $\subset \mathbf{D} \cap J$-inj.*
4. *Either $\mathbf{D} \cap I$-cof $\subset J$-cof, or $\mathbf{D} \cap J$-inj $\subset I$-inj.*

Remark. Essentially, 1. pins down the weak equivalences, and 2., 3., and 4. pin down the (trivial) cofibrations and fibrations. The proof proceeds slightly differently. Completeness and two out of three properties are inherent (in 1.); retract axioms will be automatically satisfied once we properly define the structure; 2. and 3. actually give the functorial factorisation; 4. gives the lifting axiom.

Proof. The properties are clearly satisfied by any cofibrantly generated model categories. Conversely, we define $\mathbf{D}$ to be the subcategory of weak equivalences, I-cof to be the set of cofibrations, and J-inj to be the set of fibrations. As in the remark, it suffices to check the retract axiom, lifting axiom, and functorial factorisation.

$\mathbf{D}$ is by definition closed under retracts. I-cof and J-inj are closed under retracts since retracts preserve lifting properties. We only show for left lifting properties. Say f is a retract of g, and g has left lifting property with respect to h. Then given any commutative diagram involving f and h:



where g' is the lift as a result of lifting g with respect to h. $B \to F \to C$ is the desired lift of f with respect to h.

Two types of functorial factorisation follow from the small object arguments on I and J respectively. By 6.15 and the assumptions, there is $f = \beta(f)\alpha(f) = \delta(f)\gamma(f)$, where $\alpha(f)$ is in I-cell $\subset$ I-cof $=$ the set of cofibrations, $\beta(f)$ is in I-inj $\subset \mathbf{D} \cap J$-inj $=$ the set of trivial fibrations by 3., $\gamma(f)$ is in J-cell $\subset \mathbf{D} \cap I$-cof $=$ the set of trivial cofibrations, and $\delta(f)$ is in J-inj $=$ the set of fibrations.

For the lifting axiom, we assume $\mathbf{D} \cap J$-inj $\subset I$-inj; the other case is similar, also using the retract argument and functorial factorisation for the second half of the axiom while the first half follows directly. By definition, $\mathbf{D} \cap J$-inj $\subset I$-inj implies trivial fibrations have right lifting property with respect to cofibrations i.e. cofibrations have left lifting properties with respect to trivial fibrations; this is the first half. For any trivial cofibration f, by the above we can factorise $f = \delta(f)\gamma(f)$ where $\delta(f)$ is in J-inj and $\gamma(f)$ is in J-cell. The above argument also shows $\gamma(f)$ is actually a trivial cofibrations, and by two out of three we have $\delta(f)$ is also a trivial fibration, and by the first half of the lifting axiom, f has the left lifting property with respect to this trivial fibration $\delta(f)$. Now we can use the retract argument and conclude f is a retract of $\gamma(f)$. $\gamma(f)$ is in J-cell and thus in J-cof, and since retracts preserve lifting properties, f is also in J-cof. By definition, this means f has the left lifting properties with respect to fibrations.

□

It is actually easier to verify whether an adjunction is a Quillen adjunction in the realm of cofibrantly generated model categories.

Lemma 6.20. *Suppose (F, U, ϕ) is an adjunction between a cofibrantly generated model category $\mathbf{C}$, with generating (trivial) cofibrations I and J, and a model category $\mathbf{D}$. Then it is a Quillen adjunction if and only if Ff is a cofibration for all f in I and Ff is a trivial cofibration for all f in J.*

Proof. It suffices to show the conditions are sufficient. It is not hard to show $F(I\text{-cof}) \subset FI\text{-cof}$, using adjointness and lifting properties. Let H be the class of cofibrations in $\mathbf{D}$. Then by hypothesis, $FI \subset H$, and FI-cof $\subset H$-cof. However, by 2.7, H-cof $= H$, and we have $F(I\text{-cof}) \subset H$. Thus F preserves cofibrations, and a similar argument shows that F preserves trivial cofibrations.

□

6.5 Combinatorial Model Category

A very important special case of cofibrantly generated model category is combinatorial model category, which is defined to be both cofibrantly generated and locally presentable. We will mainly present a useful judging criterion for combinatorial model categories, called Jeff Smith's theorem, and one of their most important consequences by Dugger.

To begin with, we need to define a locally presentable category, in a way that is consistent with the terminology used in this paper (see 6.4 and 6.7). To this end we choose Dugger's definition and terminology in [17]:

Definition 6.21. A category $\mathbf{C}$ is *locally presentable* if it is complete and cocomplete, and there exists a regular cardinal κ and a set of objects $\mathbf{D}$ in $\mathbf{C}$ such that

1. Every object in $\mathbf{D}$ is κ-presentable.
2. Every object in $\mathbf{C}$ can be expressed as a κ-filtered colimit of objects of $\mathbf{D}$.

We say an object A is κ-presentable if $\mathrm{Hom}_{\mathbf{C}}(A, -)$ preserves κ-filtered colimits; see [25] for the definition of a κ-filtered colimit, which is a generalisation of the λ-sequence for κ-filtered λ that we are concerned about.

It is also shown in [25] that every object X in a locally presentable category $\mathbf{C}$ is μ-presentable for some regular cardinal μ. In particular, this implies every object X is λ-small for some λ. Therefore, given a set of maps I, every domain of the maps of I is λ-small for some λ, and taking a regular cardinal that is greater than their supremum, we deduce that the domains of I are small in $\mathbf{C}$. Therefore, in a locally presentable category $\mathbf{C}$, we may run through the small object argument for any set of maps I, which is extremely useful.

Definition 6.22. A model category $\mathbf{C}$ is *combinatorial* if it is cofibrantly generated and its underlying category is locally presentable.

The guiding principle to justify a combinatorial model category is Jeff Smith's theorem.

Theorem 6.23 (Jeff Smith's Theorem)**.** *Let $\mathbf{C}$ be a locally presentable category, $\mathbf{D}$ a subcategory, and I a set of morphisms in $\mathbf{C}$. Suppose they satisfy the following:*

1. *$\mathbf{D}$ is closed under retracts and has the two out of three property.*
2. *I-inj $\subset \mathbf{D}$*
3. *$\mathbf{D} \cap I$-cof is closed under transfinite composition and pushouts.*
4. *$\mathbf{D}$ satisfies the solution set condition at I, i.e. for any morphism f in I, there exists a subset $\mathbf{D}_f$ of $\mathbf{D}$ such that any commutative square*

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle d} \\ B & \longrightarrow & D \end{array}$$

where d is in $\mathbf{D}$ can be factorised by the diagram

$$\begin{array}{ccccc} A & \longrightarrow & E & \longrightarrow & C \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle d_f} & & \downarrow{\scriptstyle d} \\ B & \longrightarrow & F & \longrightarrow & D \end{array}$$

for some d_f in $\mathbf{D}_f$.

Then letting $\mathbf{D}$ be the weak equivalences, I-cof be the cofibrations, and ($\mathbf{D} \cap I$-cof)-inj be the fibrations, one obtains a cofibrantly generated model structure on $\mathbf{C}$, making it a combinatorial model category.

We mainly follow the proof in [26], which is developed from [27] and [28], and frame it in our terminology. One already sees how 6.23 is closely related to 6.19, and indeed, the main technical gap is the following lemma. The proof of the theorem actually boils down to finding the generating trivial cofibrations J.

Lemma 6.24. *Assume $\mathbf{D} \cap I$-cof, $\mathbf{C}$, and $\mathbf{D}$ satisfy the assumptions in 6.23. Let $J \subset \mathbf{D} \cap I$-cof be a set of morphisms in $\mathbf{C}$ such that for any commutative square*

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle d} \\ B & \longrightarrow & D \end{array}$$

there exists $j \in J$ fitting into the commutative diagram

$$\begin{array}{ccccc} A & \longrightarrow & E & \longrightarrow & C \\ \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle d} \\ B & \longrightarrow & F & \longrightarrow & D \end{array}$$

Then any $f \in \mathbf{D}$ can be factorised as hg with $g \in J$-cell, $h \in I$-inj.

Moreover, J-cof $= \mathbf{D} \cap I$-cof in this case.

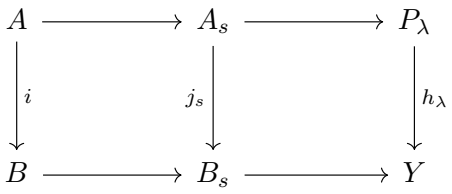
Proof of the lemma. The main part of the proof is very similar to that of the small object argument 6.15, "interpolating" the factorisation maps between. We will thus split the proof into steps, imitating our proof of 6.15.

We would like to factorise $f : X \to Y$ in $\mathbf{D}$ as $X \to P_1 \to P_2 \to \cdots \to P_\lambda \overset{h_\lambda}{\to} Y$ by transfinite induction. At a successor ordinal λ, the induction hypothesis is: there is P_λ such that $P_{\lambda-1} \to P_\lambda$ is in J-cell, and there is a map $P_\lambda \to Y$ such that the composite is f.

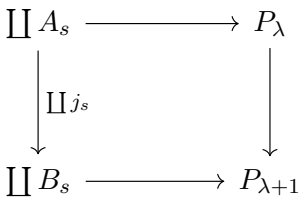
Step 1: interpolate maps in J such that $P_\lambda \to P_{\lambda+1}$ in J-cell. Let $P_0 = X$, $h_0 = f$, and assume we have built P_λ and $h_\lambda : P_\lambda \to Y$, and $P_{\alpha-1} \to P_\alpha$ is in J-cell for $\alpha < \lambda$. Then let S_λ be the set of all commutative squares such that $i \in I$

$$\begin{array}{ccc} A & \longrightarrow & P_\lambda \\ \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle h_\lambda} \\ B & \longrightarrow & Y \end{array}$$

The assumption tells us for each square there is a corresponding square:



for each $s \in S_\lambda$, since by assumption $X \to P_\lambda$ as a composition of J-cell maps is in J-cell $\subset \mathbf{D}$, and by two out of three property h_λ is also in $\mathbf{D}$. We build $P_{\lambda+1}$ as the pushout:

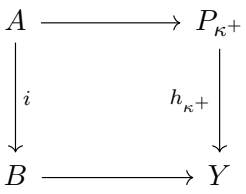


and by pushout there exists $h_{\lambda+1} : P_{\lambda+1} \to Y$. Also, $P_\lambda \to P_{\lambda+1}$ is in J-cell by 6.14. Thus the successor stage $\lambda + 1$ is done.

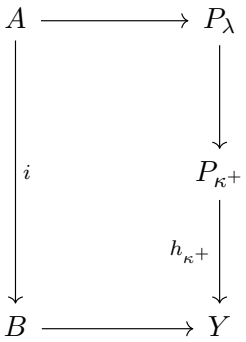
For a limit ordinal λ, $P_\lambda := \mathrm{colim}_{\alpha<\lambda} P_\alpha$. Everything is clearly satisfied.

Step 2: verify f factors through P_{κ^+} for some κ^+. Indeed, there is a regular cardinal κ such that the domains of maps in I are κ-small. By definition, κ^+ (the successor cardinal of κ) is κ-filtered. Thus we will show $X \to P_{\kappa^+} \overset{h_{\kappa^+}}{\to} Y$ is the desired factorisation. All the complicated constructions now serve to prove h_{κ^+} is in I-inj.

Given the commutative diagram

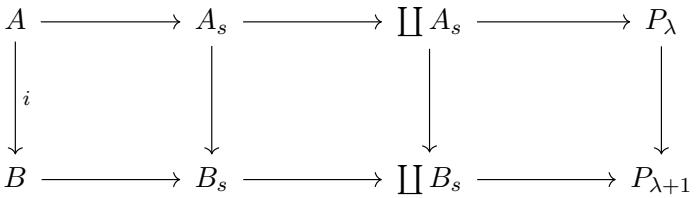


since A is κ-small and $X \to P_{\kappa^+}$ is a κ^+-sequence for κ-filtered κ^+, $X \to P_{\kappa^+}$ factors through a previous stage P_λ, and we obtain the commutative diagram



Now this lifting problem can be indexed by $s \in S_\lambda$, and by construction the diagram clearly

factors as:



thus all the constructions above give a map $B \to P_{\lambda+1}$, which then gives the desired lift $B \to P_{\lambda+1} \to P_{\kappa^+}$.

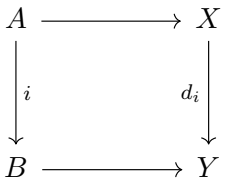
The rest is the second part.

Note that J-cof $=$ retracts of J-cell in any locally presentable model category: indeed, by small object argument and retract argument, any f in J-cof is a retraction of J-cell, and since J-cof is closed under retracts we have the reverse inclusion. See 7.16 for a similar proof.

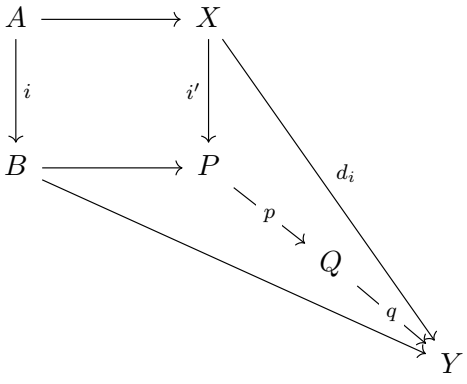
Thus, since $\mathbf{D} \cap I$-cof is closed under retracts, pushouts, and transfinite compositions, $J \subset \mathbf{D} \cap I$-cof implies J-cell $\subset \mathbf{D} \cap I$-cof. Conversely, by the above we know any f in $\mathbf{D} \cap I$-cof can be factorised as hg where $h \in I$-inj and $g \in J$-cell, so by the retract argument f is a retract of $g \in J$-cell $\subset J$-cof, and the result follows from J-cof being closed under retracts.

□

Proof of the theorem. It suffices, which we will justify in the end, to prove that there is a set of morphisms J that satisfies the properties in 6.24. Consider the set of all morphisms (in the category of arrows) from $i \in I$ to $d_i \in \mathbf{D}_f$, and then for each morphism we form a commutative diagram:

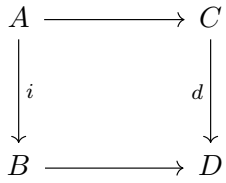


Then for such a diagram, form a pushout P:

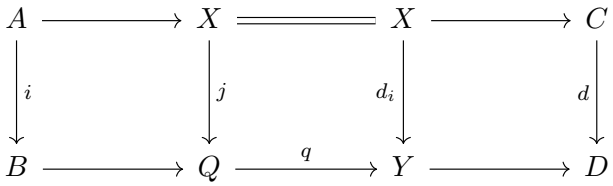


where we factorises the pushout map $P \to Y$ as $p \in I$-cell and $q \in I$-inj. Now let $j := pi'$, and we define J to be the set of such j; indeed, there is one such j for each morphism from i to d_i. Since from the diagram $i' \in I$-cell and by definition $p \in I$-cell, $j \in I$-cell$\subset I$-cof; $q \in I$-inj $\subset \mathbf{D}$ by assumption of the theorem, d_i is in $\mathbf{D}$ by definition, so by two out of three j is also in $\mathbf{D}$. Thus $J \subset \mathbf{D} \cap I$-cof.

Finally, for any diagram



the solution set condition tells us it can be factorised as



which implies that any j in J satisfies the conditions in 6.24.

Thus by 6.24, we find a set of morphisms J such that J-cof $= \mathbf{D} \cap I$-cof. Then it is clear that all conditions in 6.19 are satisfied; for example, since J-cof $= \mathbf{D} \cap I$-cof, we have J-cof $\subset I$-cof, and thus (I-cof)-inj $\subset$ (J-cof)-inj, but this is I-inj $\subset J$-inj, and the assumption I-inj $\subset \mathbf{D}$ then concludes I-inj $\subset \mathbf{D} \cap J$-inj. The rest almost follows immediately.

As a result, by 6.19, $\mathbf{C}$ is cofibrantly generated and thus combinatorial.

□

We conclude this section by a consequential result, proved by Dugger in [17].

Theorem 6.25. *Every combinatorial model category is Quillen equivalent to a simplicial model category.*

Here a simplicial model category means a model category whose hom-sets are simplicial sets, and the simplicial structure is compatible with the model structure in some specific ways; see [24, Definition 4.2.18] for details. Simplicial model categories are important since they have good simplicial structures and are in particular very close to the theory of ∞-category. For example, when $\mathbf{C}$ is a simplicial model category, its so-called underlying ∞-category is the homotopy coherent nerve of the subcategory of fibrant-cofibrant objects, which is much easier to handle. See [29, p.638] for details.

7 Examples of model categories

We will present two main examples of (cofibrantly generated) model categories, the categories of chain complexes of modules over a ring and topological spaces.

7.1 Chain complexes

We denote the category of chain complexes over a ring R by $\mathbf{Ch}(R)$. Its objects are chain complexes of R-modules with differentials d_n for $n \in \mathbb{Z}$, and its morphisms are chain maps f, namely collections of R-linear maps f_n such that $d_n f_n = f_{n-1} d_n$. Note that we will use the homological indexing convention for $\mathbf{Ch}(R)$; for a chain complex $(X_n)_{n\in\mathbb{Z}}$, we define the differential as $d_n : X_n \to X_{n-1}$.

Lemma 7.1. *Every object in $\mathbf{Ch}(R)$ is small.*

Proof. See [24, Lemma 2.3.2]. □

Let $H_n X$ be the n-th homology of a chain complex X, and let $Z_n X$ and $B_n X$ be the modules of n-cycles and n-boundaries of X, respectively. The following defines two sets of maps that will prove to be generating cofibrations and generating trivial cofibrations.

Definition 7.2. Given an R-module M, define $S^n(M) \in \mathbf{Ch}(R)$ by $S^n(M)_n = M$ and 0 otherwise; define $D^n(M)$ by $D^n(M)_n = D^n(M)_{n-1} = M$ and 0 otherwise, with $d_n = \mathrm{id}$. Let $S^n := S^n(R)$, and $D^n := D^n(R)$, and there is an obvious injection $S^{n-1}(M) \hookrightarrow D^n(M)$.

Define I to be the set of maps $S^{n-1} \to D^n$ and J to be the set of maps $0 \to D^n$. Now define the collection of cofibrations to be I-cof and fibrations to be J-inj, which we will concretely characterise later. Define a chain map f to be a weak equivalence if it is a quasi-isomorphism ($H_n(f)$ is an isomorphism for all n).

Remark. D^n and S^n draw an analogy with the disks and spheres as topological spaces. Note that $D^n, S^n : R\text{-mod} \to \mathbf{Ch}(R)$ are functors. D^n is left adjoint to the evaluation functor $\mathrm{Ev}_n : \mathbf{Ch}(R) \to R\text{-mod}$ that takes X to X_n since $\mathrm{Hom}(D^n(X), Y) \cong \mathrm{Hom}(X, \mathrm{Ev}_n(Y))$:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & X & \xrightarrow[\mathrm{id}]{} & X & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle f_n} & & \downarrow{\scriptstyle f_{n-1}} & & \downarrow{\scriptstyle 0} & & \\
\cdots & \longrightarrow & Y_{n+1} & \longrightarrow & Y_n & \xrightarrow{d_n} & Y_{n-1} & \longrightarrow & Y_{n-2} & \longrightarrow & \cdots
\end{array}$$

where the map $D^n(X) \to Y$ is completely determined by f_n since $f_{n-1} = d_n f_n$, and any f_n gives such a chain map. Similarly, S^n is left adjoint to the cycle functor $Z_n : \mathbf{Ch}(R) \to R\text{-mod}, X \mapsto Z_n X$, since $\mathrm{Hom}(S^n(X), Y) \cong \mathrm{Hom}(X, Z_n(Y))$:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle f_n} & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle 0} & & \\
\cdots & \longrightarrow & Y_{n+1} & \longrightarrow & Y_n & \xrightarrow{d_n} & Y_{n-1} & \longrightarrow & Y_{n-2} & \longrightarrow & \cdots
\end{array}$$

where $d_n f_n = 0$ i.e. the image of f_n lies in $Z_n Y$.

As a first step, we need to show the collection of weak equivalences has good properties.

Lemma 7.3. *The weak equivalences in* $\mathbf{Ch}(R)$ *are closed under retracts and have the two out of three property.*

Proof. The gist is that the homology $H_n : \mathbf{Ch}(R) \to R$-mod is a functor. The two out of three property immediately follows from that of R-linear isomorphisms. If f is a retract of g, by functoriality $H_n(f)$ is also a retract of $H_n(g)$. As in the proof of 6.17, retracts of an isomorphism are isomorphisms, and therefore f is a quasi-isomorphism if g is.

□

Next we will characterise fibrations and trivial fibrations.

Proposition 7.4. *A map* $p : X \to Y$ *in* $\mathbf{Ch}(R)$ *is a fibration if and only if* $p_n : X_n \to Y_n$ *is surjective for all* n*.*

Proof. A fibration is a map $p : X \to Y$ such that for all n there is a lift in the following commutative diagram:

$$\begin{array}{ccc} 0 & \longrightarrow & X \\ \downarrow & \overset{q}{\nearrow} & \downarrow{\scriptstyle p} \\ D^n & \longrightarrow & Y \end{array}$$

where this commutative diagram always exists since the composites are both 0. Therefore, p being a fibration is equivalent to the existence of a map $q : D^n \to X$ such that $pq = g$ for any R-linear map $g : D^n \to Y$, for any n. After going through the natural isomorphism $\mathrm{Hom}(D^n, Z) \cong \mathrm{Hom}(R, Z_n)$ in the remark, where Z is either X or Y, the maps become $q \in \mathrm{Hom}(R, X_n)$ such that $p_n q = g \in \mathrm{Hom}(R, Y_n)$. However, for R-linear maps, $\mathrm{Hom}(R, Z_n) \cong Z_n$, so this is (under the isomorphism) equivalent to the fact that for any $y \in Y_n$, there exists an element $x \in X_n$ such that $p_n(x) = y$, for all n. Namely, p_n is surjective for all n.

□

To satisfy the conditions of 6.19, we only need to show that I-inj $= \mathbf{D} \cap J$-inj and J-cof $\subset \mathbf{D} \cap I$-cof. This requires characterisations of trivial fibrations and cofibrations. I-inj $= \mathbf{D} \cap J$-inj is more straightforward:

Proposition 7.5. *A map* $p : X \to Y$ *in* $\mathbf{Ch}(R)$ *is a trivial fibration if and only if it is in* I*-inj.*

Proof. p is in I-inj if and only if there is a lift in the following commutative diagram (for all n):

$$\begin{array}{ccc} S^{n-1} & \overset{f}{\longrightarrow} & X \\ \downarrow{\scriptstyle i} & \overset{q}{\nearrow} & \downarrow{\scriptstyle p} \\ D^n & \underset{g}{\longrightarrow} & Y \end{array}$$

and as in 7.4, under isomorphisms (p is actually p_n then), such diagrams are in one-to-one correspondence with $A_n = \{(y, x) \in Y_n \oplus Z_{n-1}X : px = dy\}$. Note that x is in $Z_{n-1}X$ by the isomorphism $\mathrm{Hom}(S^n, X) \cong Z_n X$ in the remark. A lift in the diagram is an element $z \in X_n$ such that $dz = x$ and $pz = y$, where $dz = x$ comes from the fact that the map into X_{n-1} is the differential of the map into X_n.

Now suppose p is in I-inj. The natural idea is to prove that $Z_n p$, $H_n p$, and thus p are surjective, and that $H_n p$ is injective. Let $y \in Z_n Y$, then $(y, 0) \in A_n$, so there is $z \in X_n$ such that $pz = y$ and $dz = 0$, which immediately implies $Z_n p : Z_n X \to Z_n Y$ is surjective. Then $H_n p$ is surjective. Therefore, for any $y \in Y_n$, $dy \in Z_{n-1} Y$, so there is a class $x \in Z_{n-1} X$ such that $px = dy$. Then $(y, x) \in A_n$, so there is $z \in X_n$ such that $pz = y$. Then p_n is surjective for all n, and by 7.4, p is a fibration.

To prove that $H_n p$ is injective, we need its kernel to be 0. For any $x \in Z_n X$ such that $px = dy$ for some $y \in Y_{n+1}$, $(y, x) \in A_n$, and thus there is $z \in X_{n+1}$ such that $dz = x$, which means $x \in B_n X$. Then $H_n p$ is injective and thus an isomorphism, and p is a weak equivalence.

Conversely, assume p is a trivial fibration. Given any $(y, x) \in A_n$, we need to find $z \in X_n$ such that $pz = y$ and $dz = x$. Surjectivity of p gives the following short exact sequence:

$$0 \to K \to X \xrightarrow{p} Y \to 0$$

where K is the kernel of p. This induces a long exact sequence in homology:

$$\begin{array}{ccccccccc} \dots & \longrightarrow & H_{n+1}K & \longrightarrow & H_{n+1}X & = & H_{n+1}Y & & \\ & & & & & \overset{\partial}{\swarrow} & & & \\ H_nK & \xrightarrow{h} & H_nX & = & H_nY & \longrightarrow & \dots & & \end{array} \tag{3}$$

where the isomorphisms come from p being a weak equivalence. Then $\ker(\partial) = H_{n+1}Y$, $\ker(h) = \operatorname{im}(\partial) = 0$, and $\operatorname{im}(h) = 0$. Thus $H_n K$ must be 0 for all n.

Now choose $w \in X_n$ such that $pw = y$. Since $p(dw) = d(pw) = dy = px$, $dw - x \in K$, and $d(dw - x) = dx = 0$ gives $dw - x \in Z_{n-1}K$. Then $dw - x \in B_{n-1}K$, and there is $v \in K_n$ such that $dv = dw - x$. Letting $z = w - v$, we have $dz = x$ and $pz = y$.

□

Remark. Note that 3 tells us that p is a trivial fibration in $\mathbf{Ch}(R)$ if and only if it is a surjection with an acyclic kernel. One direction is shown, but the other is also apparent: if $H_* K$ is acyclic, then the map between $H_n X$ and $H_n Y$ is both injective and surjective. This is the characterisation we will frequently use later.

The next aim is to identify the cofibrant objects and cofibrations. The lifting property inherent in cofibrations already insinuates something close to projectivity. In fact, cofibrant objects are exactly the DG-projective complexes, and our next two lemmas are essentially proving one direction of this i.e. cofibrant objects are DG-projective.

Proposition 7.6. *If A is a cofibrant chain complex, then A_n is a projective R-module for all n. As a partial converse, any bounded below complex of projective R-modules is cofibrant.*

Proof. For an R-module A_{n-1}, we need to find a lift $h' : A_{n-1} \to M$ for every map $f : A_{n-1} \to N$, where there exists a surjection $q : M \twoheadrightarrow N$. Now p induces a surjection $p : D^n M \to D^n N$, which is q in degree n and $n - 1$. Its kernel is certainly acyclic since the chain map is identity between M and 0 otherwise. Thus we have a trivial fibration p. The map f clearly induces a chain map $g : A \to D^n N$:

$$\begin{array}{ccccccccc} \dots & \longrightarrow & A_n & \xrightarrow{d} & A_{n-1} & \longrightarrow & \dots & & \\ & & \downarrow{\scriptstyle fd} & & \downarrow{\scriptstyle f} & & & & \\ \dots & \longrightarrow & N & \longrightarrow & N & \longrightarrow & \dots & & \end{array}$$

and since A is cofibrant, it has left lifting property with trivial fibrations p, which gives a map $h : A \to D^n M$. Then apparently $h' = h_{n-1}$ is the desired lift of f. Hence A_{n-1} is projective.

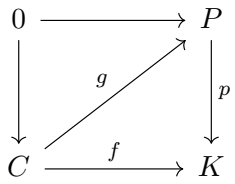
On the other hand, suppose A is a bounded below complex of projective R-modules, and $p : X \to Y$ is a trivial fibration. Then $K := \ker p$ is acyclic. Given any $g : A \to Y$, we need a lift $h : A \to X$ of g, which we will construct inductively. Boundedness provides a base case of the induction, and assume for all $k < n$, there is a map h_k such that $p_k h_k = g_k$ and $dh_k = h_{k-1}d$. Since p_n is surjective and A_n is projective, there is $f : A_n \to X_n$ such that $p_n f = g_n$. We then use the trick in the proof of 7.5 again: produce a map $F : A_n \to Z_{n-1}K$ out of f since the kernel, being acyclic, provides a preimage of d that satisfies the relation. Let $F = df - h_{n-1}d$. Then $p_{n-1}F = dg_{n-1} - g_{n-1}d = 0$ and $dF = 0$, which implies $F : A_n \to Z_{n-1}K = B_{n-1}K$. Again, since A_n is projective, lifting along $d_n : K_n \twoheadrightarrow B_{n-1}K$ gives $G : A_n \to K_n$ such that $dG = F$. Then let $h_n = f - G$, and $p_n h_n = p_n f_n = g_n$, and by definition $dh_n = df - F = h_{n-1}d$.

□

The next lemma says that $\mathrm{Hom}_{D(R)}(C, K) = 0$, where $D(R) = \mathrm{Ho}(\mathbf{Ch}(R))$. This is equivalent to saying that cofibrant complexes are actually DG-projective.

Lemma 7.7. *If C is a cofibrant chain complex, then every map from C to any acyclic chain complex K is chain homotopic to 0.*

Proof. Let P be $\mathrm{cone}(\mathrm{id}_K)[-1]$, which is the chain complex defined by $P_n = K_n \oplus K_{n+1}$ with differential $d_P(x, y) = (d_K x, x - d_K y)$. Then the natural projection $p : P \to K, (x, y) \mapsto x$ has its kernel a shifted version of K, and in particular $H_*(\ker p) = H_{*-1}K = 0$. Hence p is a trivial fibration, and cofibrant C gives the following diagram:



where $g(x) = (f(x), g_2(x))$ since $pg = f$. Now define $s = g_2 : C_n \to K_{n+1}$, which will be the desired chain homotopy. Since g is a chain map, $d_P g(x) = g d_C(x) \implies d_P(f(x), s(x)) = (f d_C(x), s d_C(x)) \implies (df_C(x), f(x) - d_K s(x)) = (f d_C(x), s d_C(x))$. Thus $f - ds = sd$, and f is chain homotopic to 0. □

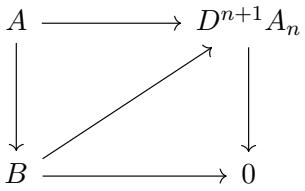
We are now prepared to characterise the cofibrations.

Proposition 7.8. *A map $i : A \to B$ in $\mathbf{Ch}(R)$ is a cofibration if and only if it is a (degreewise split) injection with cofibrant cokernel.*

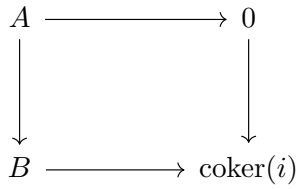
Remark. We say a chain map $i : A \to B$ is degreewise split if there exists $j_n : B_n \to A_n$ such that $j_n i_n = \mathrm{id}_A$. Let C be the cokernel of the chain map i. The word "split" comes from the fact that the short exact sequences $0 \to A_n \to B_n \to C_n \to 0$ are split, but this actually follows automatically once we have C_n to be projective. Therefore, it is actually a consequence of "injection with cofibrant cokernel" and should be redundant here.

Proof. First assume i is a cofibration. As in the remark, it suffices to show that i is an injection with cofibrant cokernel; however, to prove injectivity, the best way is to find a left inverse. There

is a natural map $A \to D^{n+1}A_n$ that is id_A in degree n. Consider the following diagram:

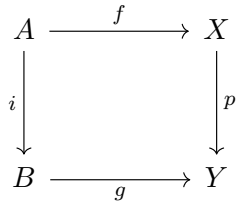


where there is a lift $B \to D^{n+1}A_n$ since $D^{n+1}A_n$, being the kernel, is acyclic. Looking at degree n, we find a left inverse of i_n and i is a split monomorphism. To show the cokernel is cofibrant, we know $\mathrm{coker}(i)$ is actually a pushout:



and the collection of maps defined by lifting properties are closed under pushouts. Then $0 \to \mathrm{coker}(i)$ is also a cofibration, as required.

Now suppose i_n is an inclusion for all n and its cokernel C is cofibrant. Then by 7.6, each C_n is projective. Given a diagram



where p is a quasi-isomorphism and degreewise surjection, we must find a lift $B \to X$. Projectivity of C_n shows the existence of the lift, and DG-projectivity ensures the lift has good properties. DG-projectivity in some sense plays the role of "acyclic kernels" in some previous proofs, for example in 7.5 and 7.6. Since C_n is projective, we may write $B_n = A_n \oplus C_n$. The differential $d : A_{n+1} \oplus C_{n+1} \to A_n \oplus C_n$ is given by $d(a,c) = d(a,0) + d(0,c) = (da, 0) + (\tau c, dc)$ for some $\tau : C_n \to A_{n-1}$, which follows from i and the cokernel map being chain maps. There might be a twisted term just because the cokernel map does not give enough information, and, since $d^2 = 0$. τ must satisfy $d\tau + \tau d = 0$. Then the map g in the diagram is defined by $g(a,c) = g(a,0) + g(0,c) = gi(a) + g(0,c) = pf(a) + \sigma(c)$ for some $\sigma : C \to Y$ such that $d\sigma = pf\tau + \sigma d$. The last condition results from $dpf(a) + d\sigma(c) = dg(a,c) = gd(a,c) = g(da + \tau c, dc) = pfd(a) + pf\tau(c) + \sigma d(c) \implies d\sigma(c) = (pf\tau + \sigma d)(c)$. For a lift h in the diagram, $h_n : A_n \oplus C_n \to X$, the equation $hi = f$ gives $h(a,c) = f(a) + \nu(c)$ for some collection of maps $\nu_n : C_n \to X_n$. Therefore, our next goal is to find $\nu_n : C_n \to X_n$ such that $p\nu = \sigma$ and $d\nu = \nu d + f\tau$.

Our strategy is similar to before: construct a map that is defined by "chain map conditions" and lands in the kernels. Since C_n is projective for all n, there are maps $G_n : C_n \to X_n$ such that $p_n G_n = \sigma_n$. Consider $r = dG - Gd - f\tau : C_n \to X_{n-1}$. Then by the condition on σ, $pr = 0$ and r is actually a map $C_n \to K_{n-1}$; moreover, $dr = -dGd + f\tau d = -rd$, so $r : C \to \Sigma K$ is a chain map, where ΣK is the suspension of K given by $(\Sigma K)_n = K_{n-1}$ and $d_{\Sigma K} = -d_K$. By 7.7, since the kernel of p is acyclic, r is chain homotopic to 0. Thus there is a map $s : C_n \to (\Sigma K)_{n+1}$ such that $-ds + sd = r$. Let $\nu = G + s$. Then $p\nu = pG = \sigma$ since s lands in the kernel, and

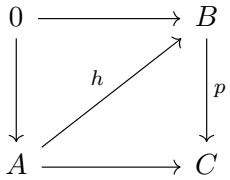
$d\nu = dG + ds = r + Gd + f\tau + sd - r = \nu d + f\tau$, as required. Therefore, $h = (f, \nu) : B \to X$ is the desired lift in the diagram.

□

It remains to show that J-cof is part of the trivial cofibrations. Remember J-cof is defined to have the left lifting properties with the fibrations, which makes it simpler to deal with.

Proposition 7.9. *A map $i : A \to B$ is in J-cof if and only if i is an injection whose cokernel is projective as a complex. In particular, every map in J-cof is a trivial cofibration.*

Remark. We first point out that projective chain complexes are cofibrant objects. Cofibrations are I-cof which has left lifting property with I-inj that is proved to be trivial fibrations in 7.5. Given any diagram



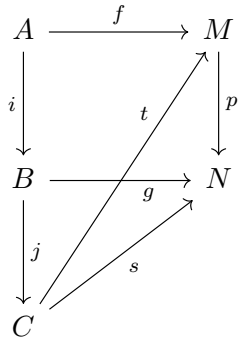
there is a lift h since p, being a fibration, is a surjection as a chain map, and A is projective as a chain complex. Thus A is indeed cofibrant. A is also acyclic, which we will explain in the proof.

Another thing to notice is that projectivity as a complex also implies the existence of a retract as a chain map. Since $\mathbf{Ch}(R)$ is also an abelian category, if there is a projective chain complex C in the following sequence:

$$0 \to A \xrightarrow{i} B \to C \to 0$$

then the sequence is also exact, and there will be a retract chain map $r : B \to A$ such that $ri = \mathrm{id}_A$.

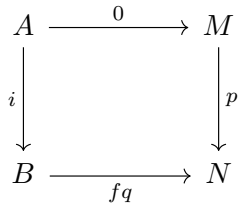
Proof. By 7.4, J-cof are the maps that have the left lifting properties with all surjections. Suppose first that i is an injection with projective cokernel $j : B \to C$ and there is a commutative diagram (the square in the following picture)



where p is surjective. By the remark, there is a retraction $r : B \to A$. Then we apply the common trick involving retractions, that the map $pfr - g : B \to N$ satisfies $(pfr - g)i = 0$ and thus factors through a map $s : C \to N$. Since C is projective, s lifts to a map t such that $pt = s$. Then $fr - tj$ is the desired lift in the diagram, and i is in J-cof.

Conversely, assume $i : A \to B$ is in J-cof. Then by definition it has left lifting property with every map in J-inj, which is the set of fibrations. This in particular implies it has left lifting property with every trivial fibration which was already shown in 7.5 to be I-inj, so J-cof $\subset$ I-cof. Then i is an injection. Let $q : B \to C$ denote the cokernel of i, and it remains to show that C

is projective. Given a map $f : C \to N$ and surjective map $p : M \to N$, we have the following diagram



There is a lift $h : B \to M$ such that $hi = 0$ and $ph = fq$. It then follows that h factors through a map $g : C \to M$ such that $gq = h$, and thus $pgq = ph = fq$. Since q is surjective, this implies $pg = f$ and g is the required lift. Then C is projective.

It remains to show the second part. As in the remark, projective cokernels must be cofibrant, so the only thing left is to prove i is a weak equivalence i.e. a quasi-isomorphism. Similar to what we have shown in the remark after 7.5, i is a weak equivalence if and only if its cokernel is acyclic, and it suffices to prove projective chain complexes must be acyclic. Let $P = \mathrm{cone}(\mathrm{id}_C)[-1]$, defined as $P_n = C_n \oplus C_{n+1}$ and $d(x, y) = (dx, x - dy)$. Then there is a surjection $P \to C$, and id_C lifts to a map $C \to P$. The second component is a collection of maps $D_n : C_n \to C_{n+1}$ such that $dD + Dd = \mathrm{id}$. Applying this to any cycle x, we have $dDx = x$, so x is also a boundary, and C is acyclic.

□

We have already finished the proof of the following theorem.

Theorem 7.10. $\mathbf{Ch}(R)$ *is a cofibrantly generated model category with* I *as its generating set of cofibrations,* J *as its generating set of trivial cofibrations. Weak equivalences are quasi-isomorphisms, fibrations are surjections, and cofibrations are injections with DG-projective cokernels.*

Proof. As we have said before, it suffices to show, by 6.19, that J-cell $\subset \mathbf{W} \cap I$-cof and I-inj $= \mathbf{W} \cap J$-inj, where $\mathbf{W}$ is the collection of quasi-isomorphisms. 7.5 shows I-inj $= \mathbf{W} \cap J$-inj, and 7.9 proves J-cof $\subset \mathbf{W} \cap I$-cof, which clearly implies J-cell $\subset \mathbf{W} \cap I$-cof. □

Remark. Another frequently used model category is $\mathbf{Ch}_{\geq 0}(R)$, the bounded below chain complexes of R-modules. Clearly, by 7.6, A in $\mathbf{Ch}_{\geq 0}(R)$ is cofibrant if and only if it is a complex of projective R-modules. Therefore, there is also a projective model structure on $\mathbf{Ch}_{\geq 0}(R)$ with weak equivalences the quasi-isomorphisms, fibrations the surjections at degree greater than 0, and cofibrations the injections with degreewise projective cokernels. Note that we only need surjections at degree $n \geq 1$ since the generating trivial fibrations can only be $0 \to D^n$ for $n \geq 1$, while $D^0 \notin \mathbf{Ch}_{\geq 0}(R)$.

We can also impose another useful model structure on $\mathbf{Ch}(R)$, called the injective model structure. We will only introduce it here and exemplify a simple Quillen equivalence. There are many more possible model structures on $\mathbf{Ch}(R)$ or the category of differential graded algebras; see [30] for a thorough account.

Theorem 7.11. *There is also an injective model structure on* $\mathbf{Ch}(R)$*, where the weak equivalences are quasi-isomorphisms, fibrations are surjections with fibrant kernels, and cofibrations are injections. Similarly, every fibrant object is a complex of injective, and every bounded above complex of injectives is fibrant.*

Remark. It is not hard to prove that $\mathrm{id} : \mathbf{Ch}(R)_{\mathrm{proj}} \to \mathbf{Ch}(R)_{\mathrm{inj}}$ is a Quillen equivalence. It is clearly a left Quillen functor: cofibrations and trivial cofibrations in $\mathbf{Ch}(R)_{\mathrm{proj}}$ are injections with some properties, and they are apparently cofibrations and trivial cofibrations in $\mathbf{Ch}(R)_{\mathrm{inj}}$,

which are just injections and injections with acyclic cokernels. One may also use 6.20. By 5.10, it remains to show id reflects weak equivalences and for any fibrant Y in $\mathbf{Ch}(R)_{\text{inj}}$, the map $FQUY \to Y$ is a weak equivalence, where U is the adjoint of id, which is also id. Both follow from definitions since $QY \to Y$ is inherently a weak equivalence.

In the end we will present a nice application of the projective model structure, related to the homotopy category of $\mathbf{Ch}(R)$ we talked about earlier. From the example one can tell how powerful a model structure is, encompassing most we need in the derived category. We are using without proofs that the path object for a chain complex X is $X_n \oplus X_{n+1} \oplus X_n$ at each degree, a fact that we will explain later in the next section.

Proposition 7.12. *Let $\mathbf{C} = \mathrm{Ho}(\mathbf{Ch}_{\geq 0}(R))$. Then for any R-module X and Y, there is a natural isomorphism*

$$\mathrm{Hom}_{\mathbf{C}}(S^m X, S^n Y) \cong \mathrm{Ext}_R^{n-m}(X, Y)$$

for any nonnegative m and n. Here $\mathrm{Ext}_R^k(X, Y) = 0$ for negative k.

Proof. The core property is that the cofibrant replacement of $S^m X$ in $\mathbf{Ch}_{\geq 0}(R)$ is the "projective resolution" of X. To find the cofibrant replacement $Q(S^m X)$ of $S^m X$, it suffices to find a cofibrant object Y such that there is a trivial fibration from $Y \to S^m X$. Let P be the complex of projective resolution of X, and define Z as $Z_k = P_{k-m}$ for $k \geq m$ and 0 otherwise. Then since the augmented projective resolution is exact, and $P_0 \xrightarrow{\epsilon} X$ is surjective, the chain map f with $f_m = \epsilon$ and 0 otherwise is indeed a fibration:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle \epsilon} & & \downarrow & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & X_m & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

where the upper complex is indeed cofibrant since at each degree it is projective.

□

Now let $Z = Q(S^m X)$. Since Z is weakly equivalent to $S^m X$, they are the same in the homotopy category, and together with 4.8, we have

$$\mathrm{Hom}_{\mathbf{C}}(S^m X, S^n Y) = \mathrm{Hom}_{\mathbf{C}}(Z, S^n Y) \cong \pi(Z, S^n Y)$$

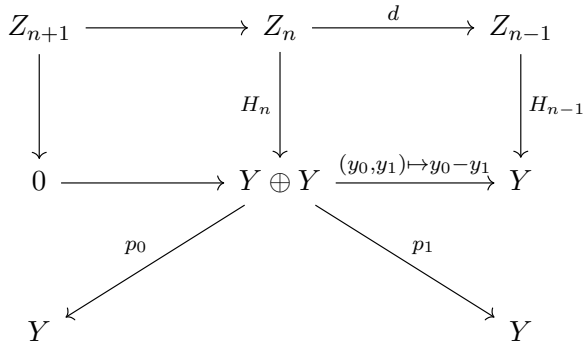
where Z is cofibrant and $S^n Y$ is fibrant. Then without loss of generality we can assume $m = 0$ since the cofibrant replacement does not have lower degree anyways. For example, if $m > n$, then Z_k is 0 for $k < m$, and $S^k Y$ is 0 for $k > m$, so the only map between them is 0. Thus we may assume $n \geq 0$.

Let A be the path object for $S^n Y$ i.e. $A_n = Y \oplus Y$, $A_{n-1} = Y$ if $n > 0$, and 0 otherwise. The path object structure maps $q : S^n Y \to A$ and $p_0, p_1 : A \to S^n Y$ are given by:

$$q : a \mapsto (a, a), \quad p_i : (y_0, y_1) \to y_i$$

By definition of $\pi(Z, S^n Y)$, two maps f, g are equal in $\pi(Z, S^n Y)$ if and only if they are related to each other by a right homotopy i.e. there exists $H : Z \to A$ such that $p_0 H = f$ and $p_1 H = g$.

When $m = 0$, Z is exactly the projective resolution of A. As before, a map $f : Z \to S^n Y$ is equivalent to a map $f_n : Z_n \to Y$ such that $fd = 0$. Then by the following part of a diagram:



where $Y \oplus Y \to Y$ is given by the canonical differential, we see that $p_0 H = f, p_1 H = g$ gives $H_n = (f_n, g_n)$, and the composite $Z_n \to Y$ is $f_n - g_n$. Therefore, f is right homotopic to g if and only if there exists $h = H_{n-1} : Z_{n-1} \to Y$ such that $hd = f_n - g_n$. But then by the Hom complex of projective resolutions:

$$\mathrm{Hom}_R(Z_{n-1}, Y) \xrightarrow{d' = -\circ d} \mathrm{Hom}_R(Z_n, Y)$$

two maps f_n, g_n are equal in $\mathrm{Ext}^n_R(X, Y)$ if and only if their difference lie in the image of d' above, which is equivalent to the existence of a map $h : Z_{n-1} \to Y$ such that $hd = f_n - g_n$.

Rigorously, we may let $\Phi : \mathrm{Hom}_{\mathbf{C}}(S^m X, S^n Y) \to \mathrm{Ext}^{n-m}_R(X, Y)$ be the map induced by $\mathrm{Hom}_{\mathbf{C}}(S^m X, S^n Y) \cong \pi(Z, S^n Y)$, and the argument above shows that Φ is both injective and surjective, thus a natural bijection.

7.2 Topological spaces

We will briefly introduce the model structure on the category of topological spaces **Top** and their basic properties, in preparation for stable model categories.

Definition 7.13. A map $f : X \to Y$ in **Top** is called a weak homotopy equivalence if the induced map $\pi_n(X, x) \to \pi_n(Y, f(x))$ is an isomorphism for all $n \geq 0$ and $x \in X$. Here at $n = 0$ it is an isomorphism of sets, and for $n > 0$, it is a group isomorphism.

A map $f : X \to Y$ is a Serre fibration if, for each CW-complex A, f has the right lifting property with respect to the inclusion $A \to A \times [0, 1]$. Equivalently, the set of Serre fibrations consists of the maps that have the right lifting property with respect to the inclusions $D^n \to D^n \times I$, where D^n are the n-disks.

Theorem 7.14. *The category of topological spaces* **Top** *can be made into a model category with weak equivalences being weak homotopy equivalences, fibrations being Serre fibrations, and cofibrations being the maps having left lifting properties with respect to acyclic fibrations.*

Precisely, **Top** *is a cofibrantly generated model category where the generating cofibrations are the boundary inclusions $S^{n-1} \to D^n$, S^n being the n-sphere, and the generating trivial cofibrations are $D^n \to D^n \times I$.*

Proof. See [24, Section 2.4]. □

Let $\mathbf{T}$ be the category of compactly generated topological spaces. Then there is actually an identical model structure on $\mathbf{T}$, see [24, Theorem 2.4.23, 2.4.25]. There is also an induced

model structure on $\mathbf{Top}_*$ and $\mathbf{T}_*$, with the maps replaced with basepoint-preserving maps. Note that we care about $\mathbf{T}$ since over $\mathbf{T}$, the function space functor $(-)^X$ is right adjoint to the product functor $- \times X$, rendering it a closed monoidal model category structure. These are summarised as follows.

Theorem 7.15. *The inclusion $\mathbf{T} \to \mathbf{Top}$ is a Quillen equivalence. There are induced model structures on $\mathbf{T}_*$ and $\mathbf{Top}_*$, which are also Quillen equivalent.*

Remark. We need to clarify the cofibrations and fibrations. As usual, the objects in I-cell are called relative cell complexes, and this corresponds to the (transfinite version of) conventional notion of cell complexes in $\mathbf{Top}$ since the pushout

$$\begin{array}{ccc} S^{n-1} & \longrightarrow & X \\ \downarrow & & \downarrow{\scriptstyle f} \\ D^n & \longrightarrow & Y \end{array}$$

is exactly attaching a cell D^n to X along the inclusion $S^{n-1} \to X$. It turns out by the next lemma that the cofibrations are retracts of relative cell complexes, and cofibrant objects are retracts of cell complexes.

Every object is fibrant since every map $f : D^n \to X$ clearly induces a map $D^n \times [0,1] \to X$ by assigning f to every slice: $g(x,t) = f(x)$.

Lemma 7.16. *In cofibrantly generated model category $\mathbf{C}$ with generating cofibrations I, I-cof = retracts of I-cell.*

Proof. We have in fact already proved the lemma. If f is a cofibration, by the small object argument $f = pi$ where f has the left lifting property with respect to the trivial fibration p in I-inj, and by retract argument f is a retract of i in I-cell. Also, I-cell $\subset$ I-cof by 6.12, and retracts of I-cof are still in I-cof since I-cof are defined by lifting properties. Thus retracts of I-cell are still in I-cof.

□

Note that every space X is weakly equivalent (weakly homotopy equivalent) to a cell complex since the small object argument implies that we have a factorisation $\emptyset \xrightarrow{i} Y \xrightarrow{p} X$ where $i \in I$-cell $p \in I$-inj; then by definition Y is a cell complex, and p is a trivial fibration. Hence in discussing homotopy problems (for example in homotopy category of $\mathbf{Top}$), we may view every space as a cell complex.

We will give one more definition, which has been used in the omitted proof that $\mathbf{Top}$ is a model category and will be used in our discussion of stable model category.

Definition 7.17. Given two pointed topological spaces (X, x_0) and (Y, y_0), their smash product $X \wedge Y := X \times Y / X \vee Y$, where $X \vee Y$ is the wedge sum of X and Y. Precisely, $X \wedge Y = X \times Y / \sim$, where $(x, y_0) \sim (x_0, y)$ for all $x \in X, y \in Y$.

Remark. For example, $S^1 \wedge S^1$ is homeomorphic to S^2 since it's essentially shrinking the 1-skeleton to a point. The cell complex then becomes a point attached with a 2-disk, which is exactly S^2. By the same token, we can prove that $S^m \wedge S^n = S^{m+n}$: $S^m \times S^n$ has one 0-cell, one m-cell, one n-cell and one $m+n$-cell, while $S^m \vee S^n$ has one 0-cell, one m-cell and one n-cell.

The smash product gives rise to the reduced suspension of two pointed space, which is useful later.

8 Stable model category

8.1 Homotopy colimits

In this section we will give a simplified account of homotopy colimits, building up to suspension and loop functors. The case of homotopy limits is similar, completely dual to the constructions we will give.

The following definition comes from [31].

Definition 8.1. We say a poset category I is *simple* if the function $d : I \to \mathbb{N} \cup \{\infty\}$ defined by $d(i) = \{\sup(n) : \text{there are non-identity maps } x_0 \to x_1 \to \cdots \to x_n = i\}$ only takes finite values for each $i \in I$.

Basically, the categories $I = (2 \leftarrow 1 \to 3)$ and $I = \mathbb{N} = (0 \to 1 \to \cdots \to n \to \dots)$ are both simple categories.

Proposition 8.2. *Let I be a simple category and $\mathbf{C}$ a model category. Then there is a model category structure on $\mathbf{C}^I$ called the projective model structure such that $f : X \to Y$ is a weak equivalence (resp. fibration) if and only if $f_i : X_i \to Y_i$ is a weak equivalence (resp. fibration) for every $i \in I$.*

Proof. See [24, Theorem 5.1.3]. □

Definition 8.3. The homotopy colimit of a diagram $X \in \mathbf{C}^I$ is given by the colimit of a cofibrant replacement of X in $\mathbf{C}^I$ with the projective model structure.

The homotopy colimit is then well-defined up to weak equivalences. Moreover, by the nature of the definition and cofibrant replacements, if $X \to Y$ is a degreewise weak equivalence for diagrams X and Y, then they are weakly equivalent in $\mathbf{C}^I$ and so are their cofibrant replacements, and by definition their homotopy colimits are also weakly equivalent. This solves the issue that the usual colimit of a diagram may not be invariant (up to weak equivalences) under weak equivalences.

The following theorem characterises concretely the homotopy colimits. It is also one of the reasons why we require I to be small: by definition $\operatorname{colim}_j \in I_{<i} X_j$ is the colimit over $I_{<i}$, and simplicity of I ensures the well-definedness of this colimit since every chain in $I_{<i}$ has finite length.

Theorem 8.4. *Assume $X \in \mathbf{C}^I$ is degreewise cofibrant for model category $\mathbf{C}$ and simple category I. If for all $i \in I$, the map*

$$\operatorname{colim}_{j \in I_{<i}} X_j \to X_i$$

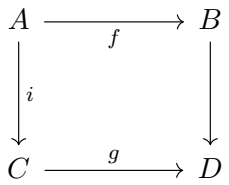
is a cofibration in $\mathbf{C}$, then X is cofibrant in $\mathbf{C}^I$.

Proof. See [32, Theorem 6.36]. □

Example 8.5. We look at the example of homotopy pushouts, and homotopy sequential colimits behave similarly. For $I = (2 \leftarrow 1 \to 3)$, assume each X_i is cofibrant. Then 8.4 asks for the following: at X_1, X_1 is automatically cofibrant; at X_2, we need $X_1 \to X_2$ to be cofibrant; at X_3, the diagram category $I_{<3}$ only consists of 1 since in the poset only $1 < 3$, and the map $X_1 \to X_3$ is cofibrant. Therefore, it suffices to replace every object with cofibrant objects and every leg with cofibrations. This is also the case in homotopy sequential colimits, and homotopy pullbacks/sequential limits.

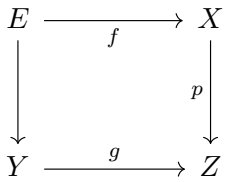
We will need the following definition.

Definition 8.6. A model category $\mathbf{C}$ is left proper if for every diagram



where f is a weak equivalence and i is a cofibration, the map g is also a weak equivalence.

Dually, $\mathbf{C}$ is right proper if for every diagram

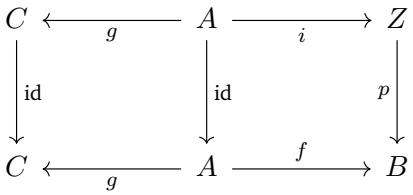


where p is a fibration and g is a weak equivalence, the map f is a weak equivalence. $\mathbf{C}$ is proper if it is both left proper and right proper.

The standard examples $\mathbf{Ch}(R)$, $\mathbf{Top}$ and $\mathbf{sSet}$ with the model structures we introduced are all proper model categories.

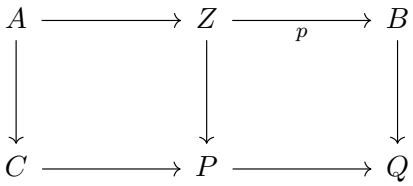
Lemma 8.7. *If $\mathbf{C}$ is a left proper model category and X is in $\mathbf{C}^I$ for $I = (2 \leftarrow 1 \rightarrow 3)$, then if just one of the legs of the diagram X is a cofibration between cofibrant objects of $\mathbf{C}$, then its pushout is its homotopy pushout.*

Proof. By functorial factorisation, we have the following diagram:



where g is by assumption a cofibration, and p is a trivial fibration, i is a cofibration by factorisation. Then the homotopy pushouts of the two rows agree since they are degreewise weakly equivalent, and the homotopy pushout of the top row is by definition its usual pushout. Thus, it remains to show the pushout of the bottom row is weakly equivalent to the pushout of the top row.

Now let P be the pushout of the top row. Then there is a diagram



where both squares are pushout squares, and thus the outer square is also a pushout. Then Q is indeed the pushout of $C \leftarrow A \rightarrow B$. By definition of left properness, $P \rightarrow Q$ is also a weak equivalence since p is. This proves the pushouts are weakly equivalent.

□

There is a dual version of this lemma that the homotopy pullbacks in a right proper model category can be computed by replacing only one leg of the pullback diagrams.

8.2 Stability

We denote the homotopy pushout of $A \leftarrow X \rightarrow B$ as $\mathrm{hpo}(A \leftarrow X \rightarrow B)$, and homotopy pullback of $A \rightarrow X \leftarrow B$ as $\mathrm{hpb}(A \rightarrow X \leftarrow B)$; meanwhile abbreviate pushouts and pullbacks as po and pb. Below we give a slightly more general version of suspension and loop functors.

Definition 8.8. Let $\mathbf{C}$ be a pointed model category with a terminal object $*$. We define endofunctors $\Sigma, \Omega : \mathrm{Ho}(\mathbf{C}) \rightarrow \mathrm{Ho}(\mathbf{C})$ as follows:

$$\Sigma(X) = \mathrm{hpo}(* \leftarrow X \rightarrow *), \quad \Omega(X) = \mathrm{hpb}(* \rightarrow X \leftarrow *)$$

In proper model categories, the suspension and loop functors are easier to deal with.

Proposition 8.9. *If $\mathbf{C}$ is a left proper model category where X is a cofibrant object in $\mathbf{C}$, $\Sigma(X) = \mathrm{po}(* \leftarrow X \coprod X \rightarrow X \wedge I)$, where $X \wedge I$ as before is the (good) cylinder object of X. Dually, if $\mathbf{C}$ is a right proper model category and X is fibrant, $\Omega X = \mathrm{pb}(* \rightarrow X \times X \leftarrow X^I)$.*

Proof. We will only prove the statement for Σ, and the case of Ω is similar. Naturally one would like to replace one leg $X \rightarrow *$ by some cofibrations between cofibrant objects, which leads to finding an object related to X yet weakly equivalent to a point $*$. A standard choice would be the cone of X as in $\mathbf{Ch}(R)$ or $\mathbf{Top}$, which we will define rigorously as the pushout of the following diagram:

$$\begin{array}{ccc} X & \xrightarrow{i_0} & X \wedge I \\ \downarrow & & \downarrow \\ * & \xrightarrow{h} & CX \end{array}$$

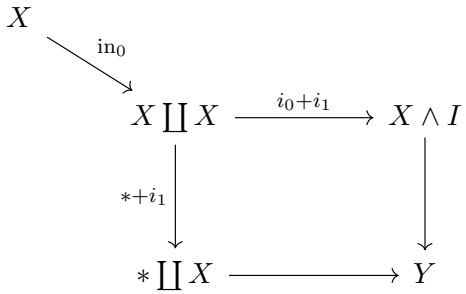
Recall that as in 3.1, $X \coprod X \overset{i_0+i_1}{\rightarrow} X \wedge I$ is the map from $X \coprod X$ to a very good cylinder object $X \wedge I$ of X, and for $p : X \wedge I \overset{\sim}{\rightarrow} X$, $pi_0 = \mathrm{id}_X$, $pi_1 = \mathrm{id}_X$. Intuitively, CX is collapsing one end of the cylinder $X \wedge I$, like the cone in the space. It is then natural to define $j : X \overset{i_1}{\rightarrow} X \wedge I \overset{h}{\rightarrow} CX$ i.e. $j = hi_1$. This actually gives us a factorisation $X \overset{j}{\rightarrow} CX \rightarrow *$, where we want to show j is a cofibration and $CX \rightarrow *$ is a weak equivalence.

We first prove j is a cofibration. Consider the pushout square

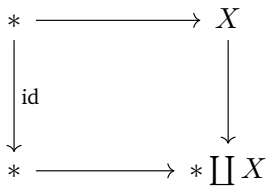
$$\begin{array}{ccc} X \coprod X & \xrightarrow{i_0+i_1} & X \wedge I \\ \downarrow{\scriptstyle *+i_1} & & \downarrow \\ * \coprod X & \longrightarrow & \bullet \end{array}$$

where $X \coprod X \rightarrow * \coprod X$ is the map induced by sending the first copy of X to $*$ and keeping the

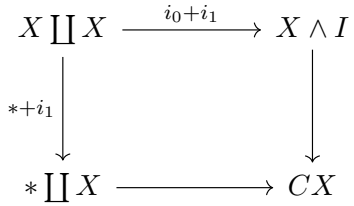
second copy. For any Y completing this diagram, we can form another diagram



where as in 3.1. in_0 is mapping the first copy of X into $X \coprod X$. Then the two routes into Y induce a commutative diagram in the definition of CX, which gives a map $CX \to Y$ commuting with all the maps (commuting with the second copy is immediate). Thus CX is actually a pushout of the diagram. Now $X \to * \coprod X$ is a cofibration since there is a pushout diagram

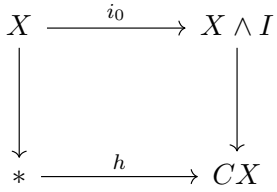


By observing the pushout diagram we just proved:



we see $* \coprod X \to CX$ is a cofibration by 2.7, and $j : X \to * \coprod X \to CX$ is exactly the composite of the two maps and is thus also a cofibration. Since X is a cofibrant object, this shows CX is also cofibrant.

Then we prove $CX \to *$ is a weak equivalence. By 3.2, i_0 is an acyclic cofibration, and by the pushout square



h is also an acyclic cofibration, so by two out of three and $* \to CX \to *$ being a weak equivalence, $CX \to *$ is also a weak equivalence.

As a result, by 8.7, by replacing $X \to *$ with $X \to CX$ where $j : X \to CX$ is a cofibration, $*$ is weakly equivalent to CX, and both X and CX are cofibrant, we have $\mathrm{hpo}(* \leftarrow X \to *) = \mathrm{po}(* \leftarrow X \to CX) = \mathrm{po}(* \leftarrow X \to \mathrm{po}(* \leftarrow X \to X \wedge I))$. Then by the "associativity" formula for pushouts i.e. $\mathrm{po}(A \leftarrow B \to \mathrm{po}(C \leftarrow D \to E)) = \mathrm{po}(A \coprod C \leftarrow B \coprod D \to E)$, taking $A = C = *, B = D = X, E = X \wedge I$, we have $\mathrm{hpo}(* \leftarrow X \to *) = \mathrm{po}(* \leftarrow X \coprod X \to X \wedge I)$. Note that the previous identity of CX can also be proved in this way, which seems like a detour.

The dual case is similar, using the object that is a pullback of $(* \to X \leftarrow X^I)$.

□

Example 8.10. We will compute Σ and Ω of $\mathbf{Ch}(R)$ in this example. Since Σ and Ω are functors on $\mathrm{Ho}(\mathbf{C})$, it suffices to calculate them for cofibrant or fibrant objects in $\mathbf{C}$; for a general object X in $\mathbf{C}$, we can take its cofibrant or fibrant replacement and then apply Σ or Ω, and the result is identical up to isomorphisms in $\mathrm{Ho}(\mathbf{C})$.

For cofibrant X in $\mathbf{Ch}(R)$, $X \wedge I$ is the usual cylinder of X, defined by $Y_n = X_n \oplus X_{n-1} \oplus X_n$, and X^I is the object P defined by $P_n = X_n \oplus X_{n+1} \oplus X_n$. Note that the cone CX is the usual mapping cone $(CX)_n = X_n \oplus X_{n-1}$ with its differential, after collapsing one end of Y. The structure maps are given by:

$$X \oplus X \xrightarrow{(a,b)\to(a,0,b)} Y \xrightarrow{(a,b,c)\to a+c} X$$

Then we have a pushout square

$$\begin{array}{ccc} X \oplus X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \Sigma(X) \end{array}$$

which is by definition the cokernel of $(a, b) \mapsto (a, 0, b)$. This is $Y/\mathrm{im}(X \oplus X)$, and $(Y/\mathrm{im}(X \oplus X))_n = (X_n \oplus X_{n-1} \oplus X_n)/(X_n \oplus X_n) = X_{n-1} = X[-1]_n$. Thus $\Sigma(X) = X[-1]$.

Similarly, $\Omega(X) = \ker(X^I \to X \oplus X) = X[1]$.

Example 8.11. We are now computing the suspension and loop functors in the category of pointed compactly generated weak Hausdorff topological spaces $\mathbf{T}_*$ (the functors are only defined for pointed model category; for general pointed topological spaces $\mathbf{Top}_*$, some of the calculations below should work, but Σ and Ω do not form an adjunction).

Let $I_+ = I \cup \{\mathbf{pt}\}$, and let $X \wedge I_+$ be the smash product of X with I_+, as defined in 7.17. The cylinder object of X is exactly $X \wedge I_+$, which we call the reduced cylinder of X. By abuse of notation, denote the path object of X as X^I, which by definition is the space of paths in X, i.e. the set of all continuous maps $[0, 1] \to X$, and the basepoint of X^I is considered as the constant path at $*$. The maps are given by:

$$X \coprod X \xrightarrow{(x,y)\mapsto[x.0]\cup[y,1]} X \wedge I_+ \xrightarrow{[x,t]\mapsto x} X$$

$$X \xrightarrow{x\mapsto([0,1]\mapsto x)} X^I \xrightarrow{\gamma\mapsto(\gamma(0),\gamma(1))} X \times X$$

Here $X \wedge I_+ = X \times I_+/X \vee I_+ = X \times I/\{*\} \times I$, which identifies a vertical line of the cylinder as the basepoint. Thus the structure maps for cylinder objects are basically identifying X with the faces of the cylinder, and the maps for path objects are essentially identifying points in X with the constant paths.

Now the suspension functor satisfies the commutative diagram:

$$\begin{array}{ccc} X \coprod X & \longrightarrow & X \wedge I_+ \\ \downarrow & & \downarrow \\ * & \longrightarrow & \Sigma(X) \end{array}$$

Intuitively, this is identifying the two boundary faces as one point $*$ by definition of the structure maps. Similarly, we may view $\Sigma(X)$ as the cokernel: $\Sigma(X) = X \wedge I_+/(X \times \{0\} \cup X \times \{1\}) =$

$X \times I/(X \times \{0\} \cup X \times \{1\} \cup \{*\} \times I) \simeq X \times S^1/X \vee S^1 = X \wedge S^1$. The homotopy can be explicitly given, but intuitively it is the same as welding a reduced cylinder top by end and forming a torus of X, identifying one section of the torus and a loop inside it. In model category language, we have determined $\Sigma(X)$ up to weak equivalence, so $\Sigma(X) = X \wedge S^1$. $\Omega(X)$ is more straightforward, satisfying the following pullback diagram:

$$\begin{array}{ccc} \Omega(X) & \longrightarrow & X^I \\ \downarrow & & \downarrow \\ * & \longrightarrow & X \times X \end{array}$$

From the structure maps, it follows immediately that the kernel of the map $X^I \to X \times X$ consists of all the paths with $*$ being their endpoints i.e. all the loops at $*$. Hence $\Omega(X)$ is the loop space at $*$.

Definition 8.12 (Stable model category)**.** We say a pointed model category $\mathbf{C}$ is *stable* if the suspension and loop functors on $\mathbf{C}$ are equivalences of categories from $\mathrm{Ho}(\mathbf{C}) \to \mathrm{Ho}(\mathbf{C})$.

Remark. $\mathbf{Ch}(R)$ is stable since clearly $X \mapsto X[-1]$ and $X \mapsto X[1]$ are inverses of each other, and thus Σ and Ω are equivalences of categories.

On the other hand, $\mathbf{T}_*$ is not stable. The following is a brief explanation, which also illustrates the usefulness of the language of the model categories. We will use the n-th homotopy group π_n. By definition, for a pointed space $(X, *)$,

$$\pi_n(X, *) = \{f : (I^n, \partial I_n) \to (X, *)\}/\sim$$

where f is a map of pairs, $I = [0, 1]$, and $\sim$ is the usual homotopy between maps relative to the boundary i.e. the homotopy H is constant at $*$ on the boundary.

However, the map of pairs f is equivalent to $f : I^n/\partial I^n \to X$ for pointed spaces since f takes ∂I^n to a single point $*$. Now $I^n/\partial I^n \cong D^n/\partial D^n \cong S^n$. Therefore, $\pi_n(X)$ is exactly $\pi(S^n, X)$, where $\pi(A, X)$ means the homotopy classes of maps that appear in 4.4, by definition 3.3 and definition of π. By the remark after 7.15, S^n is cofibrant since it is a cell complex, and X is fibrant since every object is fibrant. Thus by 4.8, $\pi(S^n, X) \cong \mathrm{Hom}_{\mathrm{Ho}(\mathbf{C})}(S^n, X)$. If $\mathbf{Top}*$ is stable, then Σ would be an equivalence of categories on $\mathrm{Ho}(\mathbf{C})$, and in particular it would be fully faithful, inducing an isomorphism between $\pi(S^n, X)$ and $\pi(\Sigma(S^n), \Sigma(X))$.

To calculate $\Sigma(S^n)$, note the following topological facts: since $S^n \cong D^n/\partial D^n$,

$$\begin{aligned} \Sigma(S^n) &= \frac{(D^n/\partial D^n) \times I}{(D^n/\partial D^n) \times \{0, 1\} \cup \partial D^n \times I} \\ &= \frac{D^n \times I}{\partial D^n \times I \cup D^n \times \partial I} \\ &= \frac{D^n \times I}{\partial (D^n \times I)} = D^{n+1}/\partial D^{n+1} = S^{n+1} \end{aligned}$$

Thus our calculations show that if Σ is an equivalence, then $\pi_n(X) = \pi_{n+1}(\Sigma(X))$. However, $0{=}\pi_2(S^1) \neq \pi_3(S^2) = \mathbb{Z}$. Hence $\mathbf{T}_*$ is not a stable model category. See [31] for more details.

The following theorem concludes our discussion of stable model categories. We assume the knowledge of a triangulated category; see [33] for details.

Theorem 8.13. *Let* $\mathbf{C}$ *be a stable pointed model category. Then* $\mathrm{Ho}(\mathbf{C})$ *is triangulated.*

Proof. See [9, Prop 3.2.8].

□

There is also an ∞-category version of this theorem; see [34, Theorem 1.1.2.14]. The two proofs are similar, and they are related in a nice way (model category version on $\mathbf{C}$ and the ∞-category version on the underlying ∞-category of $\mathbf{C}$) by 6.25 if $\mathbf{C}$ is a combinatorial model category.

9 Conclusions

This paper has presented a coherent account of cofibrantly generated, combinatorial, and stable model categories and some of their concrete applications. In particular, it has given detailed proofs of the small object argument 6.15 and Jeff Smith's theorem 6.23, and the proposition 8.9 about how to compute Σ and Ω.

A natural direction for future work is to develop the ∞-categorical perspective, which has received relatively little attention in this paper. For example, one may ask how to identify explicitly the ∞-categorical localisation of a model category such as $\mathbf{Ch}(R)$, or how Dugger's constructions and Dwyer–Kan localisation can be reformulated in the language of ∞-categories, following the approaches of [29], [35], and [18]. There may be deeper connections between the theories of model categories and ∞-categories. Many results in the two settings are strikingly similar, although their formulations and proofs can look quite different. Moving between these perspectives may lead to significant conceptual and practical insights.

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